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(54) **PROCESS FOR THE PRODUCTION OF A
TECHNICAL ENZYME COMPOSITION
WITH LOW VISCOSITY PRODUCED BY A
FILAMENTOUS FUNGUS**

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(57) **ABSTRACT**

The present invention relates to a process for the production of a technical enzyme composition with low viscosity, a genetically modified filamentous fungus cell suitable for production of the technical enzyme composition, the use of such a genetically modified filamentous fungus cell for the production of the technical enzyme composition with low viscosity and a technical enzyme composition with low viscosity produced by such a process.

Specification includes a Sequence Listing.

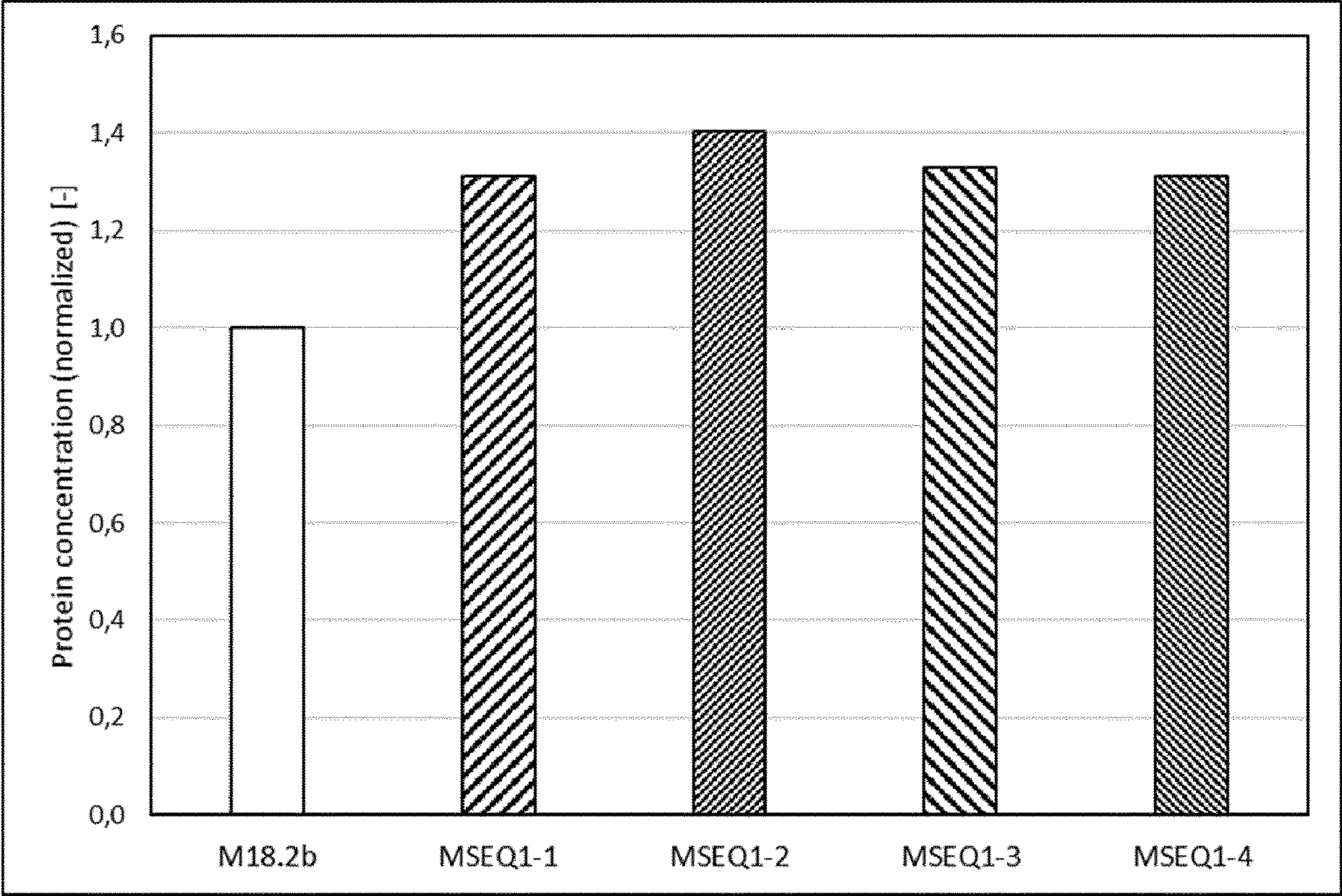


Figure 1:

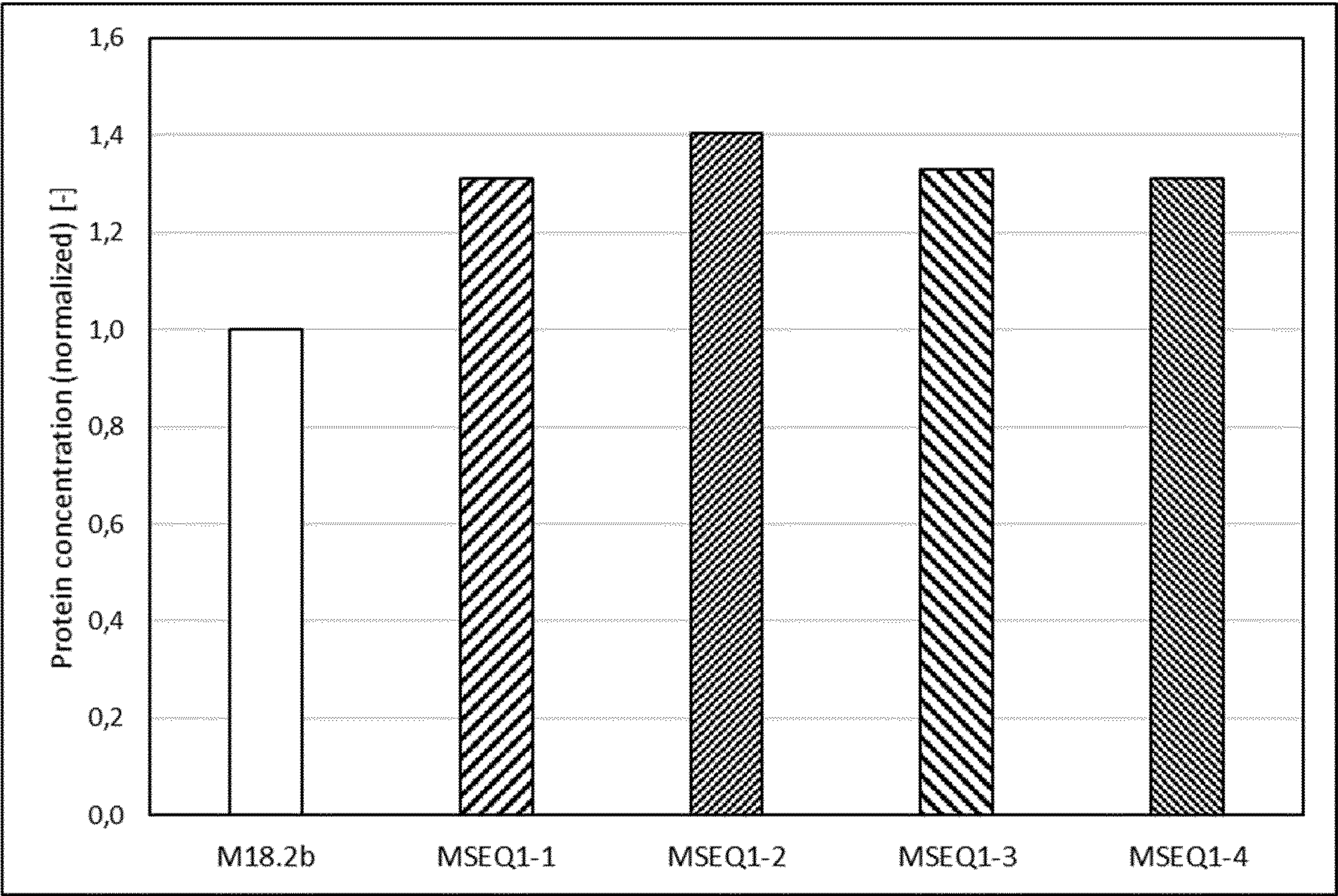


Figure 2:

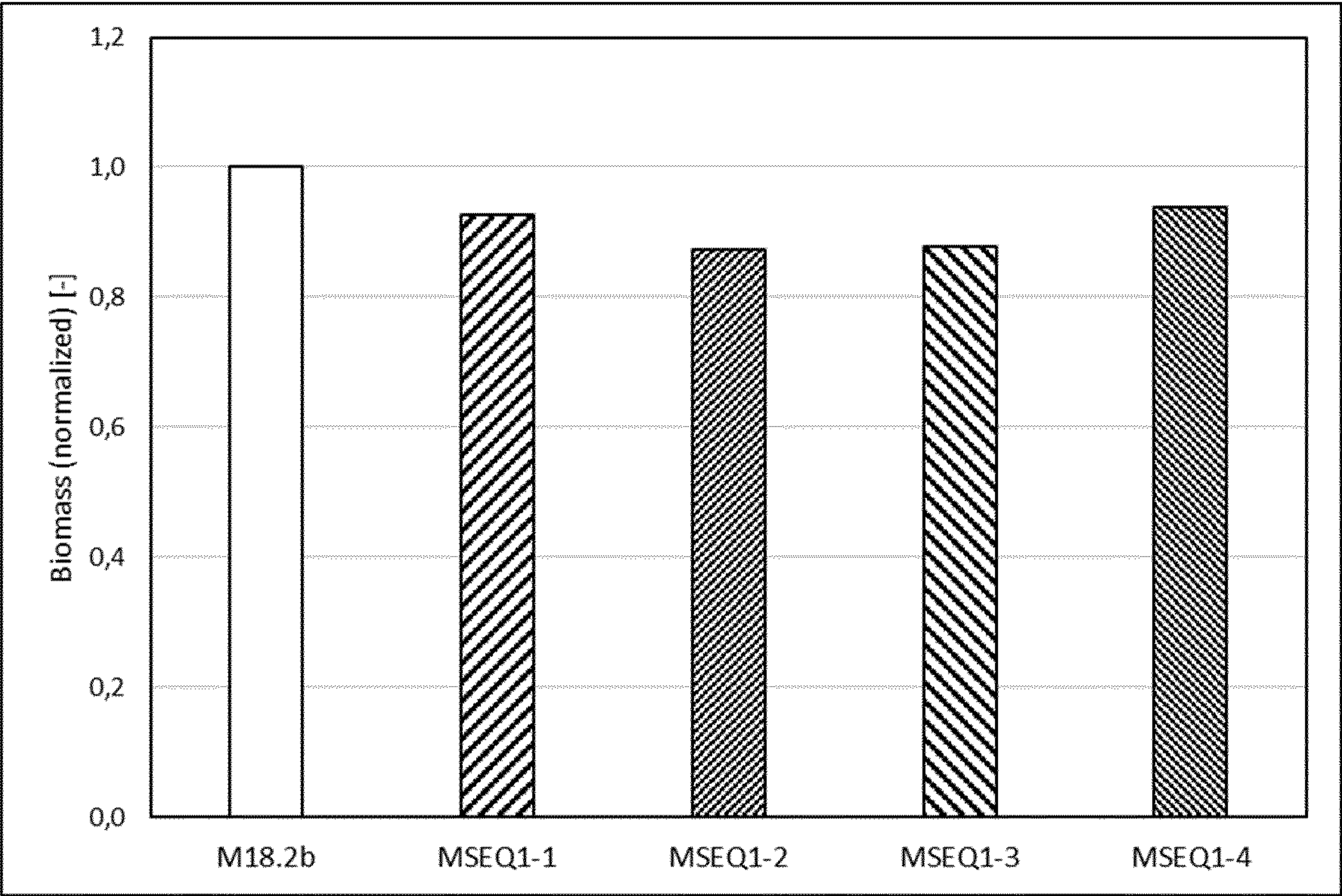


Figure 3:

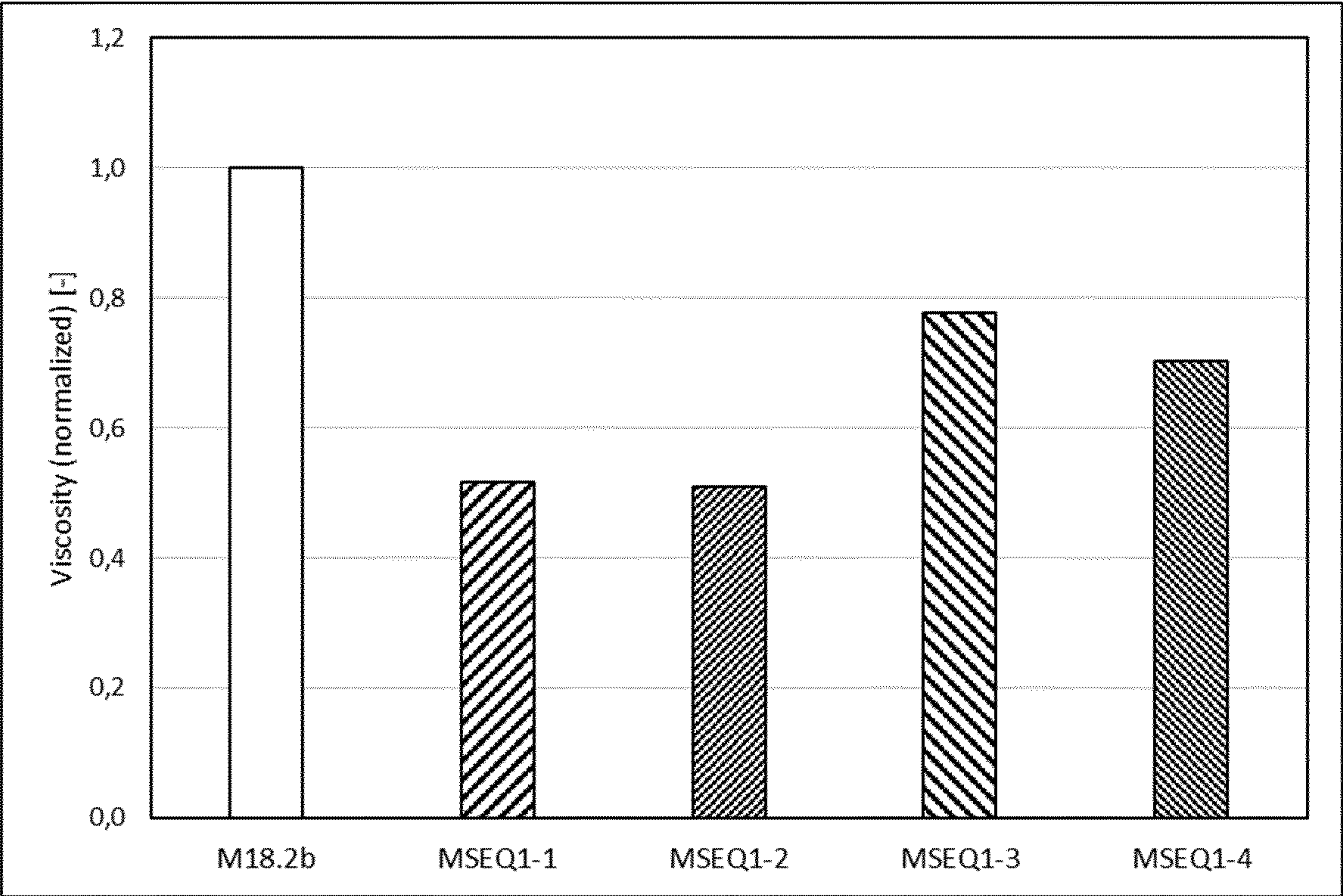


Figure 4:

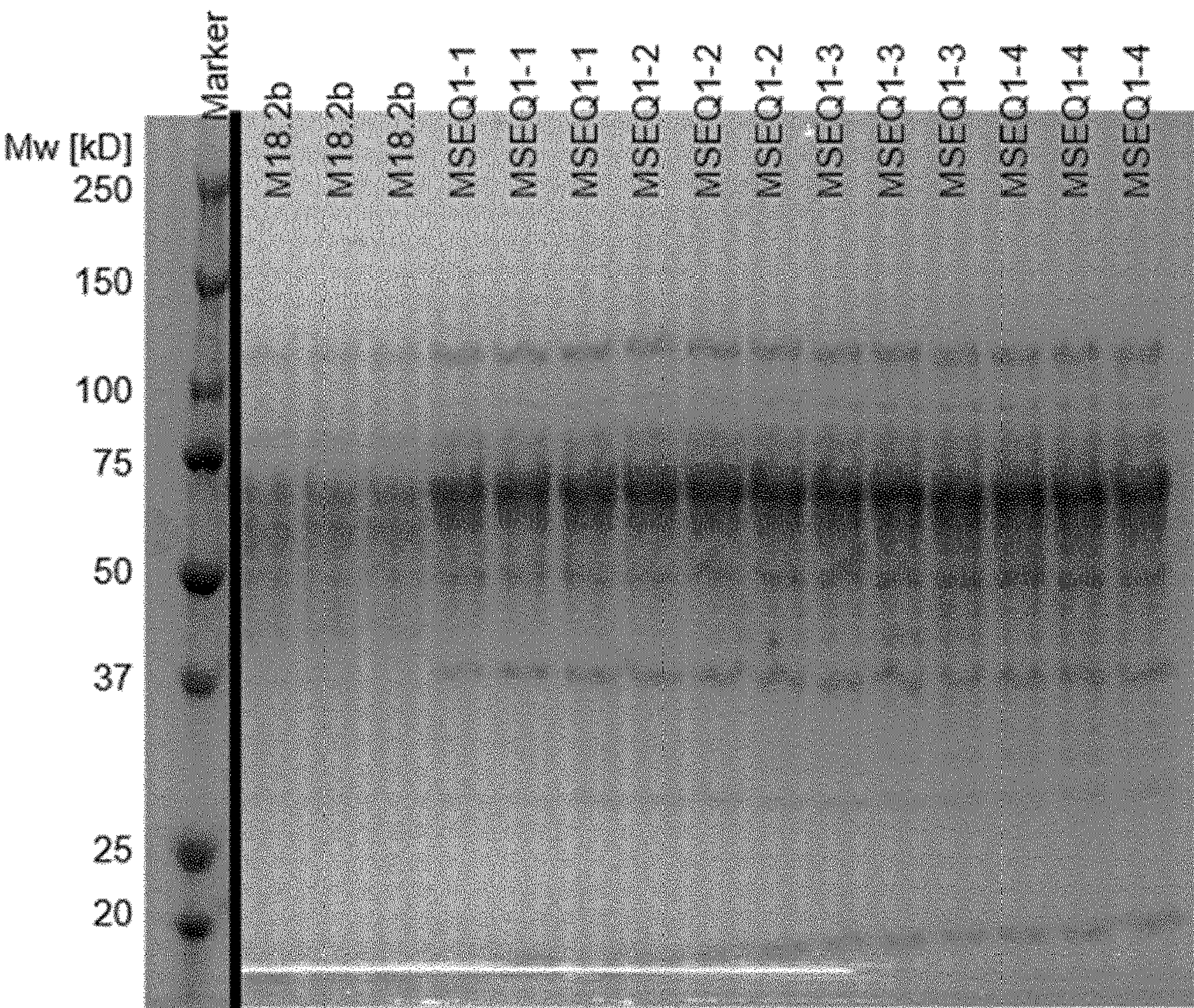


Figure 5:

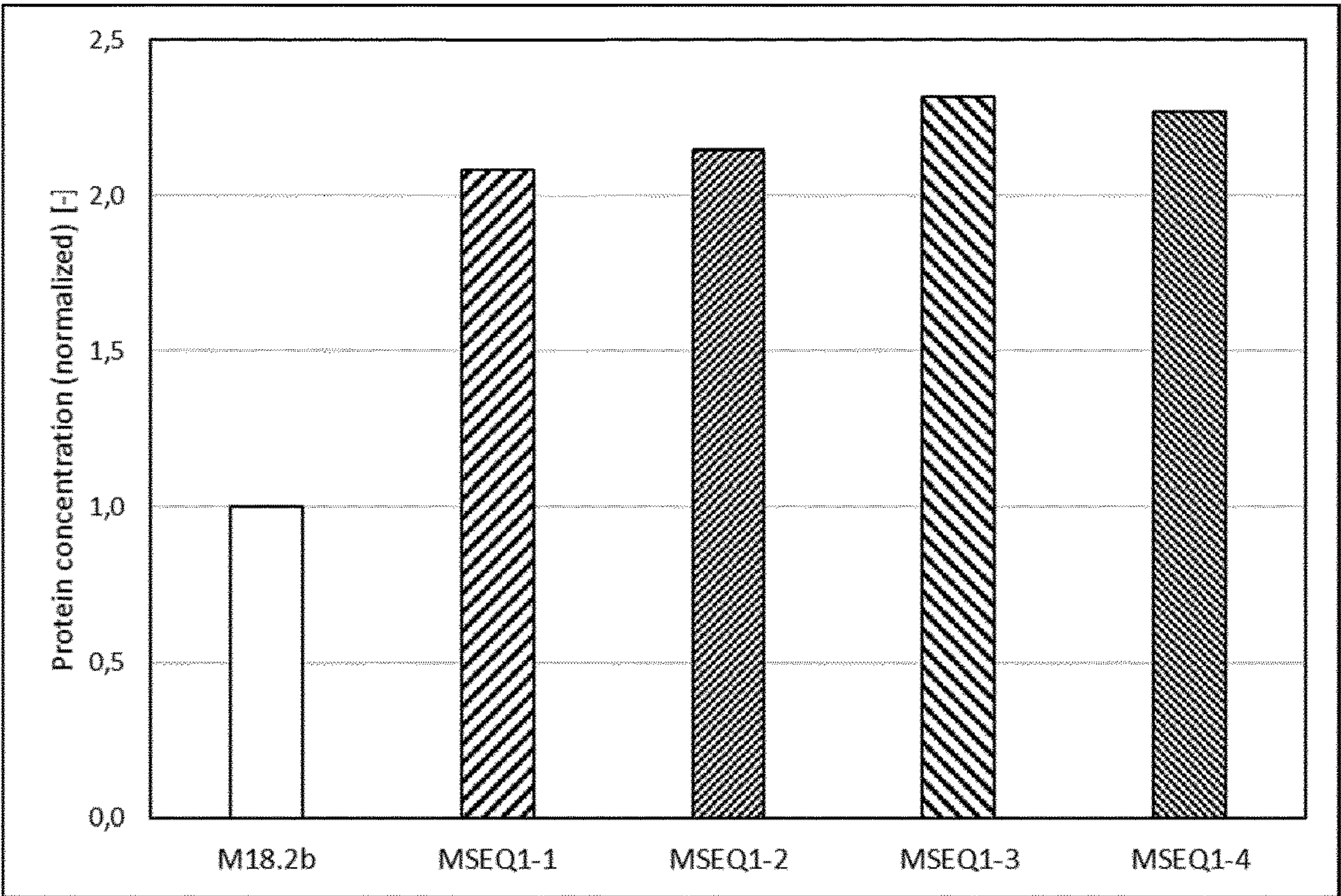


Figure 6:

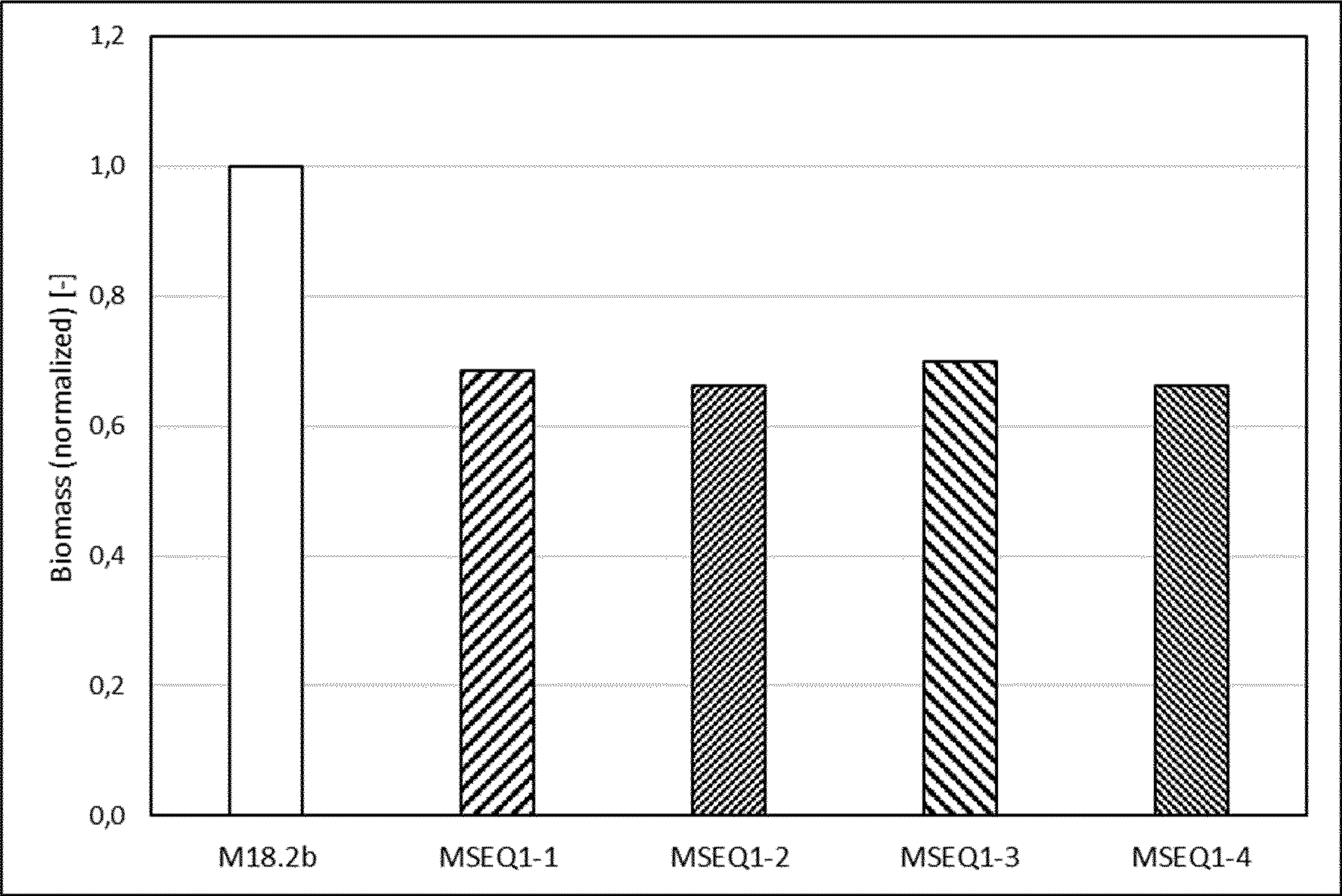


Figure 7:

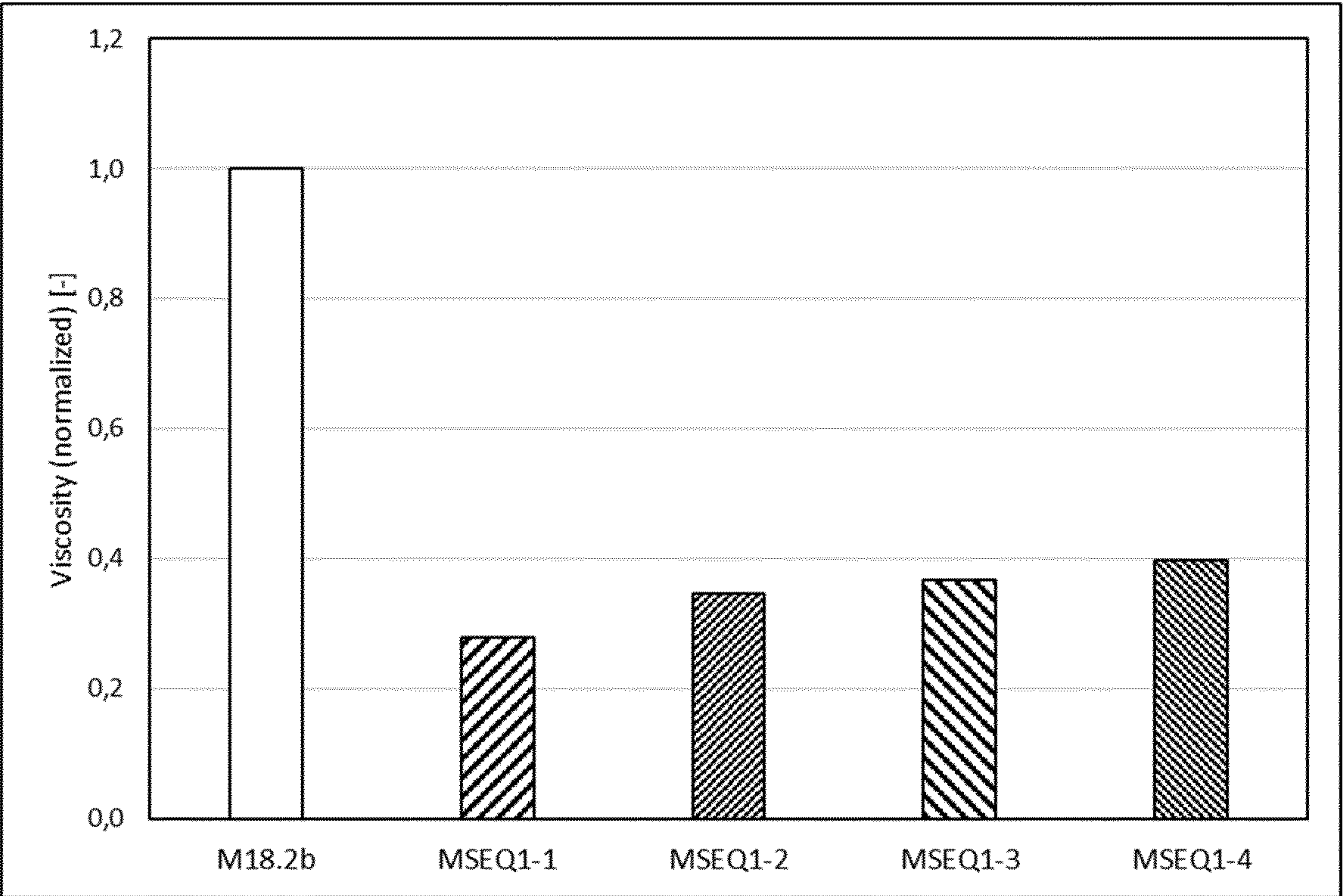
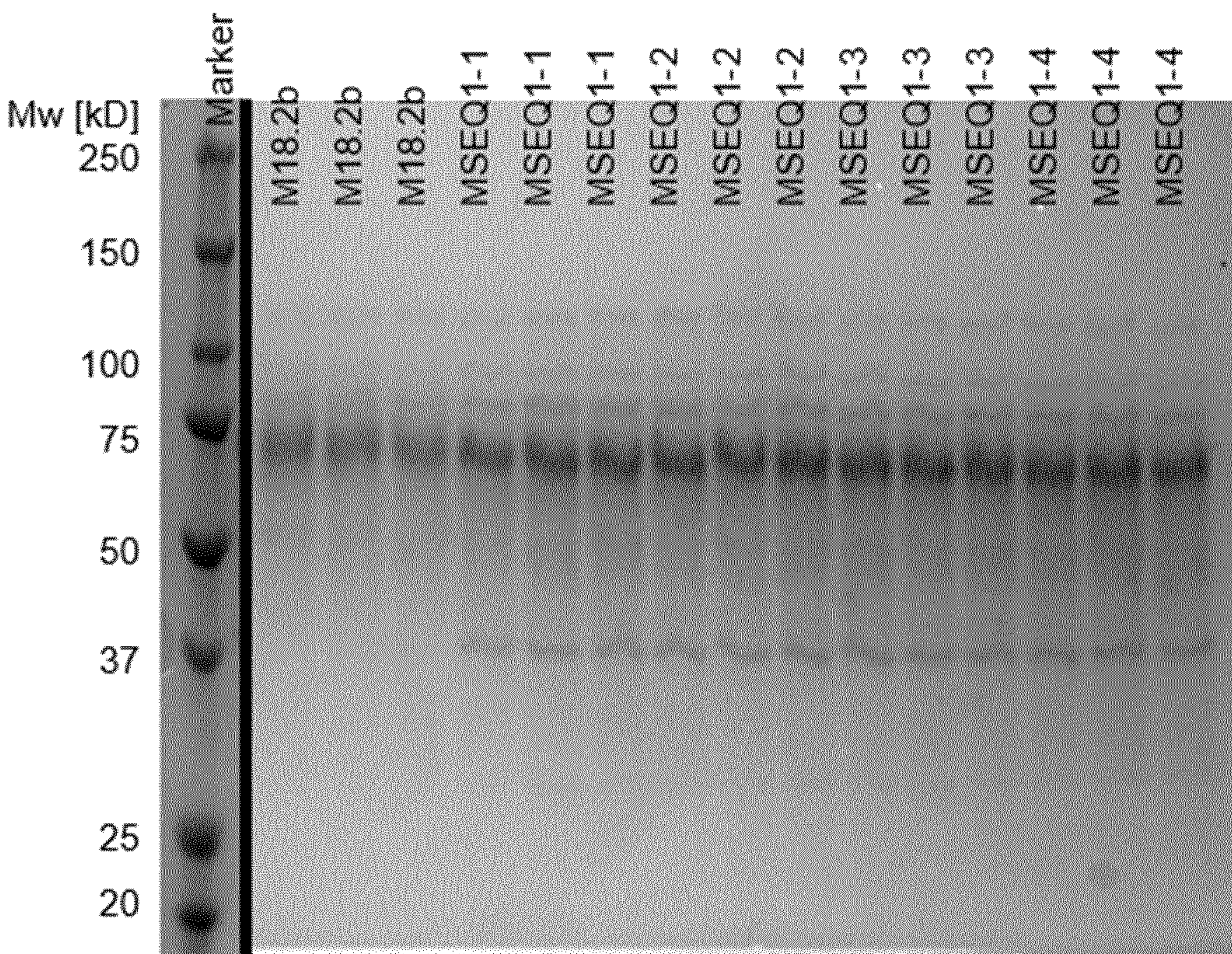


Figure 8:



**PROCESS FOR THE PRODUCTION OF A
TECHNICAL ENZYME COMPOSITION
WITH LOW VISCOSITY PRODUCED BY A
FILAMENTOUS FUNGUS**

[0001] The present invention relates to a process for the production of a technical enzyme composition with low viscosity produced by a genetically modified filamentous fungus cell, a genetically modified filamentous fungus cell suitable for production of the technical enzyme composition, the use of such a genetically modified filamentous fungus cell for the production of the technical enzyme composition with low viscosity and a technical enzyme composition with low viscosity produced by such a process.

[0002] Enzymes are important components of many commercial products and respective production processes. Modern laundry compositions contain a wide variety of different enzymes such as cellulases, many feed products for livestock contain enzymes and enzymes are also used for the production of many commercial products such as the production of bioethanol, of plastic alternatives/biodegradable plastics or even food products. Enzymes used in such processes are often called “industrial enzymes” or “technical enzymes”.

[0003] To attain economic feasibility of the desired end product, a high yield and low production cost of the used technical enzyme(s) is a necessity. This applies in particular when the desired commercial end product is a bulk product which has to compete with low price alternatives originating from cheap mineral-oil derived chemical synthesis processes.

[0004] Filamentous fungi are well known as effective producers of a wide variety of technically feasible enzymes. In addition, filamentous fungi are able to grow on a diverse range of substrates.

[0005] However, the implementation of filamentous fungi for the production of technical enzymes is still not very popular as the high viscosity of the fermentation broth of such fungi often affords time and cost consuming measures leading to too high production costs of the technical enzyme composition. In order to obtain a high yield of enzymes, a strong growth of the fungus is desired, however, strong growth results in a high content of fungus biomass within the fermentation broth. Fungi, which are known to consist of i.a. hyphae are known within the art as rendering any fermentation substrate into a high-viscous composition. This effect is significantly more distinct when a filamentous fungus is used which exhibits a sponge-like, slimy appearance.

[0006] High viscosity causes many problems, as the fungus needs constant oxygen supply by aeration during growth. In addition, cooling of the fermenter, especially in industrial-scale production is required. Both can only be guaranteed by constant stirring—on the one hand to distribute the air bubbles homogeneously within the broth, and on the other hand to facilitate constant heat-exchange with the cooling devices. The higher the viscosity of the broth the more energy needs to be spent to realize effective stirring within the reactor. Further, more air has to be pressed into the reactor causing also higher energy consumption within the compressor and sterile-filter unit. Thus, both CAPEX and OPEX increase with increasing viscosity of the fermentation broth. An alternative measure—less cell mass production—is also not attractive for commercial production as this would always be accompanied by a lower yield of technical enzyme production.

[0007] The inventors of the present invention have therefore set themselves the task to develop a process for the production of a technical enzyme composition with low viscosity produced by a filamentous fungus while maintaining a high yield of enzymes.

[0008] The task has been solved by a process for production of a technical enzyme composition, comprising the following steps:

[0009] (a) providing a fermentation medium with a glucose content of from 5 to 450 g/L;

[0010] (b) addition of at least one filamentous fungus cell wherein SEQ ID NO: 1 has been disrupted;

[0011] (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.;

[0012] (d) obtaining a technical enzyme composition.

[0013] It is of particular advantage of the inventive process that a high yield of target enzymes is achieved with any kind of medium which contains a significant amount of glucose. The majority of the proteins secreted by filamentous fungi are enzymes that degrade naturally occurring polymers such as cellulose and hemicellulose and the availability of glucose would usually prevent the filamentous fungus from producing such enzymes as they are not needed for metabolism of glucose. Further, no addition of expensive inducing substances such as gluco-oligosaccharides or sophorose is necessary. Therefore, a wide variety of different fermentation substrates which are readily and cheaply available may be used.

[0014] Within the present invention the term “technical enzyme composition” is to be understood to consist of or to contain a partly or completely fermented medium and may even contain components of the original medium but also any compound generated during the fermentation process such as enzymes. A “technical enzyme composition” may also contain part of or all of the microbial biomass of the fermentation microorganism i.e. the filamentous fungus.

[0015] Within the present invention the technical enzyme composition preferably contains at least one enzyme belonging to the class of hydrolases and/or at least one enzyme belonging to the class of oxidoreductases. Within a particularly preferred embodiment of the present invention, the technical enzyme composition contains at least one enzyme belonging to the class of hydrolases and/or at least one enzyme belonging to the class of oxidoreductases which has been produced by the at least one filamentous fungus cell. Within another also particularly preferred embodiment, the technical enzyme composition contains at least one enzyme belonging to the class of cellulases and/or at least one enzyme belonging to the class of hemicellulases which has been produced by the at least one filamentous fungus cell.

[0016] Within the present invention, the term “enzyme belonging to the class of hydrolases” is to be understood as comprising any enzyme, capable of the hydrolysis of a chemical bond. Enzymes belonging to the class of hydrolases are classified as EC 3 in the EC number classification of enzymes. According to the present invention, the term “hydrolases” comprises cellulases, hemicellulases and may also encompass pectinases, oxidases, chitinases, chitosanases, transglutaminases, pentosanases, niringinases, limoninases, lactonases, nucleases, ureases, lipoxigenases, esterases, alpha-glucanases, phosphatases, isomerases, proteases and accessory proteins.

[0017] Within the present invention, the “enzyme belonging to the class of hydrolases” may be a native enzyme of the filamentous fungus or a heterologous enzyme originating from a different species of microorganism, in particular from a different species of filamentous fungus but may also originate from a non-filamentous fungus or a bacterium.

[0018] As used within the present invention, the term “cellulase” refers to any enzyme capable of hydrolyzing cellulose polymers to shorter oligomers and/or glucose. Cellulases preferred within the technical enzyme composition include cellobiohydrolases (CBH) (EC 3.2.1.-), endo-1,4- β -glucanases (EG) (EC 3.2.1.4.), beta-glucosidase (EC 3.2.1.4), cellobiose hydrolase (EC 3.2.1.21), glycoside hydrolase 61 (GH61 and CBM33).

[0019] As used within the present invention, the term “hemicellulase” refers to any enzyme capable of degrading or supporting the degradation of hemicellulose. Hemicellulases preferred within the technical enzyme composition include β -glucanases (EC 3.2.1.-), endo-xylanases (EC 3.2.1.8), β -xylosidases (EC 3.2.1.37), acetylxyloxyesterase (EC 3.1.1.72), acetylgalactan esterase (3.1.1.6), acetyl mannan esterase, feruloyl esterase (EC 3.1.1.73), glucuronoyl esterase (EC 3.1.1.-), α -L-arabinofuranosidase (EC 3.2.1.55), α -arabinopyranosidase (3.2.1.-), α -galactosidase (EC 3.2.1.22), β -galactosidase (EC 3.2.1.23), α -glucuronidases (EC 3.2.1.139), β -mannase (EC 3.2.1.78), β -mannosidases (EC 3.2.1.25), mannan 1,4-mannobiosidase (EC 3.2.1.100), arabinogalactan endo-beta-1,4-galactanase (EC 3.2.1.89), endo-beta-1,3-galactanase (EC 3.2.1.90), galactan endo-beta-1,3-galactanase (EC 3.2.1.181), glucuronarabinoxylan endo-1,4-beta-xylanase (EC 3.2.1.136), alpha-L-fucosidase (EC 3.2.1.51), coniferin beta-glucosidase (EC 3.2.1.126), xyloglucan hydrolases (EC 3.2.1.150, 151, 155), xylan α -1,2-glucuronidase (EC 3.2.1.131), endo-xylogalacturonan hydrolase (EC 3.2.1.-; GH28), α -amylase (EC 3.2.1.1), glucan 1,4- α -glucosidase (EC 3.2.1.3), galactan 1,3-galactosidase (GH43), -1,4,-endogalactanase (EC 3.5.1.89; GH53), α -rhamnosidase (EC 3.2.1.40) and β -rhamnosidase (EC 3.2.1.43).

[0020] As used within the present invention, the term “pectinase” refers to any enzyme capable of degrading or supporting the degradation of pectin. Pectinases preferred within the technical enzyme composition include polygalacturonases (EC 3.2.1.15, 67, 82; GH28 pectin methyl esterase (EC 3.1.1.11), pectin acetyl esterase (EC 3.1.1.-), rhamnogalacturonase (EC 3.2.1.-; GH28), rhamnogalacturonan acylesterase (EC 3.1.1.86), rhamnogalacturonan galacturonohydrolase (EC 3.2.1.-), xylogalacturonan hydrolase (EC 3.2.1.-), pectin methylesterase (EC 3.1.1.11), beta-arabinofuranosidase (EC 3.2.1.55), beta-1,4-galactanase (EC 3.2.1.89), beta-1,3-galactanase (EC 3.2.1.90), beta-galactosidase (EC 3.2.1.23), alpha-galactosidase (EC 3.2.1.22), feruloyl acetyl esterase (EC 3.1.1.-), alpha-fucosidase (EC 3.2.1.51), (beta-fucosidase) (EC 3.2.1.38), beta-apiosidase (EC 3.2.1.-), alpha-rhamnosidase (EC 3.2.1.40), beta-rhamnosidase (EC 3.2.1.43), alpha-arabinopyranosidase (EC 3.2.1.-), beta-glucuronidase (EC 3.2.1.31), alpha-glucuronidase (EC 3.2.1.139), beta-xylosidase (EC 3.2.1.37) and alpha-xylosidase (EC 3.2.1.x).

[0021] As used within the present invention the term “accessory protein” refers to any enzyme capable of supporting cellulolytic enzyme activity. The term is well known to a person skilled in the art. Preferred accessory proteins

within the technical enzyme composition include Expansin, Swollenin, Loosenin and CIP Proteins (EC 3.1.1.-; CE15).

[0022] As used within the present invention, the term “oxidoreductase” refers to any enzyme capable of catalyzing an oxidation and/or a reduction reaction. Enzymes belonging to the class of oxidoreductases are classified as EC 1 in the EC number classification of enzymes. Oxidoreductase enzymes preferred within the technical enzyme composition include lytic polysaccharide monooxygenase (LPMO) (AA9-11; previously GH61 and CBM33, resp.) (EC 1.14.99.53-56, 1.14.99.B10), lignin peroxidase (EC 1.11.1.14), manganese peroxidase (EC 1.11.1.13), aryl-alcohol oxidase (EC 1.1.3.7), glyoxal oxidase (EC 1.1.3.-), carbohydrate oxidases (EC 1.1.3.4, 9, 10), cellobiose dehydrogenase (EC 1.1.99.18), catalase (hydrogen-peroxide oxidoreductase) (EC 1.11.1.6 or EC 1.11.1.21), dye-decolorizing peroxidase (EC 1.11.1.19), laccase (EC 1.10.3.2), peroxidase (EC 1.11.1.x) and versatile peroxidase (EC 1.11.1.16).

[0023] As used within the present invention, the term “esterases” refers to any enzyme capable of cleaving an ester bond. Esterases preferred within the technical enzyme composition include acetyl esterases, glucuronoyl esterases, feruloyl esterases, lipases, cutinases and phospholipases.

[0024] As used within the present invention, the term “alpha-glucanases” refers to any enzyme capable of degrading alpha-linked oligo- and polysaccharides. Alpha-glucanases preferred within the technical enzyme composition include alpha-amylases, glucoamylases, pullulanases, dextranases, trehalases, lactases, invertases and maltases.

[0025] As used within the present invention, the term “phosphatase” refers to any enzyme capable of cleaving phosphoester bonds. Phosphatases preferred within the technical enzyme composition include phytases.

[0026] As used within the present invention, the term “isomerases” refers to any enzyme capable of transferring a chemical compound into an isomeric structure. Isomerases preferred within the technical enzyme composition include xylose isomerases, glucose isomerases and arabinose isomerases.

[0027] As used within the present invention, the term “proteases” refers to any enzyme capable of cleaving a peptide bond. Proteases preferred within the technical enzyme composition include serine proteases, threonine proteases, aspartic proteases, cysteine proteases, glutamic proteases and metalloproteases.

[0028] The enzymes referenced within the present invention are classified according nomenclatures that are either based on the International Union of Biochemistry and Molecular Biology’s Enzyme Nomenclature and Classification (<http://www.chem.qmul.ac.uk/iubmb/enzyme/>) or on Carbohydrate-Active EnZYmes (<http://www.cazy.org/>) database.

[0029] According to the present invention the term “fermentation medium” is to be understood as referring to any fermentation medium known to a person skilled in the art as suitable for the inventive process. Within the process of the present invention, the fermentation medium contains from 5 to 550 g/L glucose, wherein glucose contents from 5 to 450 g/L, from 5 to 420 g/L, from 8 to 400 g/L and from 10 to 280 g/L are preferred. Further preferred ranges of glucose are from 10 to 450 g/L, from 40 to 400 g/L and from 50 to 350 g/L. Also preferred ranges of glucose are from 5 to 50 g/L, from 6 to 40 g/L or from 7 to 35 g/L and from 50 to 450 g/L, from 80 to 400 g/L and from 100 to 380 g/L. The glucose

contained in the fermentation medium may originate from any source known to a person skilled in the art as suitable for the inventive process. Within a preferred embodiment, the glucose originates from corn, sugar cane or sugar beets, preferred sources are corn syrup, sugar cane or sugar beet molasses and mixtures thereof.

[0030] Within a preferred embodiment of the present invention the “fermentation medium” can at least partly originate from chemical, mechanical and/or enzymatic hydrolysis of lignocellulosic biomass and preferably comprises prior mechanical and/or acidic pretreatment of the lignocellulosic biomass. The fermentation medium originating from chemical, mechanical and/or enzymatic hydrolysis of lignocellulosic biomass may be used “as it is” or additional glucose can be added to the fermentation medium to obtain a desired total glucose content of the fermentation medium of from 5 to 550 g/L. Glucose contents from 5 to 450 g/L glucose, 5 to 420 g/L, from 8 to 400 g/L and from 10 to 280 g/L are also suitable for the inventive process. Further preferred ranges of glucose are from 10 to 450 g/L, from 40 to 400 g/L and from 50 to 350 g/L. Also preferred ranges of glucose are from 5 to 50 g/L, from 6 to 40 g/L or from 7 to 35 g/L and from 50 to 450 g/L, from 80 to 400 g/L and from 100 to 380 g/L.

[0031] The hydrolysis of the lignocellulosic biomass has been carried out by mechanical and enzymatic hydrolysis or by sole enzymatic hydrolysis without the addition of any organic and/or inorganic acid(s). The hydrolysis of lignocellulosic biomass is known to a person skilled in the art, exemplary methods are for example described within Vishnu et al. 2012 (Trends in bioconversion of lignocellulose: Biofuels, platform chemicals & biorefinery concept in bioconversion of lignocellulose: Biofuels, platform chemicals & biorefinery concept. Progress in Energy and Combustion Science, August 2012, vol. 38 (4), 522-550) and Prasad et al. 2019 (Bioethanol production from waste lignocelluloses: A review on microbial degradation potential Chemosphere Volume 231, September 2019, p. 588-60).

[0032] Within the present invention the term “lignocellulosic biomass” is to be understood to comprise all kind of biomass known to a person skilled in the art as comprising lignocellulose. Particularly preferred lignocellulosic biomass according to the present invention includes wood, cereal straw such as but not limited to wheat straw, rice straw, barley straw, rye straw and oat straw, and/or husks and/or brans thereof, bagasse, oat hulls, switch grass, cellulose, raw paper pulp (obtained from pulp and paper production) and mixtures thereof. Additional components may comprise one or more of the following components: purified cellulose, pulp, milk whey or molasses. Lignocellulosic biomass which is particularly suitable for hydrolysis according to the process of the present invention is selected from the group consisting of cereal straw, cereal bran, cereal husks, wood, bagasse and mixtures thereof.

[0033] In a preferred embodiment the lignocellulosic biomass contains at least 25 wt.-%, preferably at least 40 wt.-%, more preferably at least 70 wt.-%, even more preferably at least 80 wt.-% and most preferred at least 90 wt.-% lignocellulose. It is to be understood that the lignocellulosic biomass may also comprise other compounds such as proteinaceous material, starch, sugars, such as fermentable sugars and/or non-fermentable sugars.

[0034] The fermentation medium originating from hydrolysis of lignocellulosic biomass has a high density of from

0.90 to 2.00 kg/L, preferably of from 0.95 to 1.90 kg/L, further preferred of from 1.00 to 1.50 kg/L and most preferred of from 1.05 to 1.35 kg/L.

[0035] The fermentation medium originating from hydrolysis of lignocellulosic biomass has a dry matter content of from 10 to 75 wt.-%, preferably of from 10 to 70 wt.-%, further preferred of from 20 to 65 wt.-%, from 30 to 65 wt.-% or from 40 to 60 wt.-% whereas a dry matter content of from 10 to 20 wt.-% and from 10 to 15 wt.-% is also preferred.

[0036] Within a preferred embodiment of the present invention, the fermentation medium further contains xylose and wherein the glucose to xylose ratio is selected from the range of from 1 to 3.5, such as a ratio selected from the range of from 1 to 3, from 1 to 2.8, of from 1 to 2.5 or of from 1 to 2.2. Further preferred ratios are 2.1, 2.0, 1.9 and 1.8.

[0037] Within an alternative preferred embodiment of the present invention, the fermentation medium further contains lactose and wherein the glucose to lactose ratio is selected from the range of from 1 to 10, such as a ratio selected from the range of from 1 to 9, from 1 to 8.5, of from 1 to 8 or of from 1 to 7. Further preferred ratios are 3, 4, 5 and 6.

[0038] Within a preferred embodiment of the present invention no gluco-oligosaccharides have been added to the fermentation medium and it is particularly preferred that the fermentation medium is free from gluco-oligosaccharides.

[0039] Within a preferred embodiment of the present invention no sophorose has been added to the fermentation medium and it is particularly preferred that the fermentation medium is free from sophorose.

[0040] Within another preferred embodiment of the present invention the fermentation medium contains less than 100 g/L cellulose and/or hemicellulose, preferably less than 80 g/L, more preferred less than 70 g/L, even more preferred less than 60 g/L, particularly preferred less than 50 g/L, and most preferred less than 40 g/L cellulose and/or hemicellulose. Within another preferred embodiment the fermentation medium of the present invention is free from hemicellulose. Within a further preferred embodiment of the present invention the cellulose content of the fermentation medium is selected from the range of from 0.01 g/L to 50 g/L, preferably from 0.1 to 40 g/L, further preferred of from 1 to 30 g/L and most preferred of from 1 to 20 g/L.

[0041] Within another preferred embodiment the fermentation medium has a nitrogen content of from 0.05 to 50.0 g/L. Preferred contents of nitrogen are selected from the range of from 0.1 to 45 g/L, from 0.3 to 40 g/L or from 0.5 to 30 g/L. In case of small scale fermentations with a total volume of fermentation medium of less than 100 L (liters), such as from 0.1 to less than 100 L, the nitrogen content of the fermentation medium is preferably selected from the range of from 0.05 to 2 g/L, further preferred of from 0.3 to 1.2 g/L and most preferred of from 0.5 to 1.0 g/L. In a preferred embodiment, small scale fermentations are carried out in reactors which are not stirred and not aerated. In case of large scale fermentations with a total volume of the fermentation medium of at least 100 L, such as for example from 100 to 10000000 L, the nitrogen content of the fermentation medium is preferably selected from the range of from 2.0 to 50 g/L, further preferred of from 5.0 to 40 g/L and most preferred of from 7.5 to 15.0 g/L. In a preferred embodiment, large scale fermentations are carried out in reactors which are stirred and/or aerated, The nitrogen can be added in any form known to a person skilled in the art as

suitable for the inventive purpose and may be added in form of ammonium sulfate, ammonia, urea, or in form of a complex nitrogen source such as soy meal, corn steep liquor, brewer's spent grains, wet distillers grains (WDG), dried distillers grains with solubles (DDGS), peptone, yeast extract or combinations thereof. In case a complex nitrogen source is used, the amount of the complex nitrogen source needed has to be calculated in alignment with the desired nitrogen content of the fermentation medium. The amount of nitrogen can be added by feeding or by adding the total amount to the fermentation medium at any time before or during step (a) and/or (b) of the inventive process. It is thereby preferred that the nitrogen is added as a 25% (wt./wt.) solution of ammonia or a 40% (wt./wt.) solution of urea.

[0042] Within another preferred embodiment of the present invention, the fermentation medium contains from 0.5 to 80 wt.-% molasses, corn syrup or mixtures thereof, preferably from 5 to 75 wt.-%, from 15 to 70 wt.-%, from 25 to 65 wt.-%, from 35 to 60 wt.-% from 38 to 55 wt.-% or from 40 to 52 wt.-%.

[0043] Within a preferred embodiment of the inventive process the pH of the fermentation medium has been adjusted to a pH selected from the range of from pH 2.0 to pH 6.0, wherein ranges of from pH 3.0 to 5.5 and from pH 3.5 to 5.5 as well as from pH 3.5 to 5.0 are particularly preferred. The adjusting of the pH can be carried out by any means and method known to a person skilled in the art as suitable for the inventive purpose. Within the process of the present invention the pH is preferably adjusted by addition of an acid such as sulfuric acid or acetic acid, NaOH, H₃PO₄ or ammonia.

[0044] Within a preferred embodiment of the inventive process the fermentation medium has a potassium hydrogen phosphate content of from 0.5 to 10.0 g/L, a magnesium sulfate heptahydrate content of from 0.05 to 1 g/L, a calcium chloride dihydrate content of from 0.1 to 1 g/L, an ammonium sulfate content of from 1.5 to 4.5 g/L, an iron (II) sulfate heptahydrate content of from 0.005 to 0.1 g/L, a manganese sulfate content of from 0.00001 to 0.001 g/L, a zinc sulfate heptahydrate content of from 0.001 to 0.01 g/L and/or a copper sulfate pentahydrate content of from 0.0001 to 0.001 g/L. Further preferred ranges are potassium hydrogen phosphate content of from 1 to 8.0 g/L, a magnesium sulfate heptahydrate content of from 0.1 to 0.8 g/L, a calcium chloride dihydrate content of from 0.3 to 0.8 g/L, an ammonium sulfate content of from 1.7 to 4.0 g/L, an iron (II) sulfate heptahydrate content of from 0.01 to 0.9 g/L, a manganese sulfate content of from 0.0001 to 0.0008 g/L, a zinc sulfate heptahydrate content of from 0.002 to 0.008 g/L and/or a copper sulfate pentahydrate content of from 0.0002 to 0.008 g/L.

[0045] The "providing" of the fermentation medium according to step (a) of the inventive process can be carried out by any method and within any means known to a person skilled in the art as suitable for the inventive process. Within a preferred embodiment the fermentation medium is provided within a batch or fed batch reactor which is preferred equipped with a stirring device and a cooling device.

[0046] According to step (b) of the inventive process, at least one filamentous fungus cell wherein SEQ ID NO: 1 has been disrupted is added to the fermentation medium. The addition of the at least one filamentous fungus cell can be carried out by any means and measure known to a person

skilled in the art as suitable for the inventive process. Within a preferred embodiment, the at least one filamentous fungus cell is added in a quantity of from 10² to 10¹⁰ cells, preferably in a quantity of from 10³ to 10⁸ cells and most preferred in a quantity of from 10⁴ to 10⁷ cells per g of fermentation medium. The at least one filamentous fungus cell can thereby be added in dried form, as conidia or in form of a preculture, containing rest of preculturing medium. It is also possible to add the at least one filamentous fungus cell in form of a fully cultured medium (also referred to as main culture).

[0047] Within the present invention the term "filamentous fungus cell" is to be understood as any cell from any filamentous fungus existing in nature and/or known to a person skilled in the art. The term also comprises any filamentous fungus cell either of natural origin or modified. The term "modified" refers to genetically and non-genetically modified fungi. i.e. fungi which have been modified by genetic methods (e.g. transformation) and non-genetic methods e.g. chemical mutagenesis or irradiation, both of which are known to those skilled in the art. Within a preferred embodiment the at least one filamentous fungus cell is selected from the group consisting of *Acremonium*, *Aspergillus*, *Chaetomium*, *Emericella*, *Fusarium*, *Humicola*, *Hypocrea*, *Irpex*, *Magnaporthe*, *Myceliophthora*, *Neurospora*, *Penicillium*, *Rhizopus*, *Talaromyces*, *Trichoderma* and *Trametes*, wherein *Trichoderma* and *Aspergillus* are particularly preferred, most preferred is *Trichoderma reesei* (teleomorph: *Hypocrea jecornia*).

[0048] It is another advantage of the present invention that in case the filamentous fungal cell is from the species *Trichoderma*, the *Trichoderma* cell produces an increased amount of at least one aspartate protease. Aspartate proteases play a significant role in breaking down complex nitrogen sources such as soy meal, corn steep liquor, brewer's spent grains, wet distillers grains (WDG), dried distillers grains with solubles (DDGS), yeast extract or peptone. Therefore, a high amount of aspartate protease(s) will enable the *Trichoderma* fungus to grow faster due to an increased availability of complex nitrogen compounds and to produce a higher amount of the technical enzyme composition within the production time. Further, a higher amount of those by-products or waste products can be incorporated into the growth medium contributing to the sustainability of the inventive process. Within the state of the art no or only a limited amount could be used as a nitrogen source and further nitrogen had to be supplemented in form of chemically synthesized ammonia or urea. A suitable growth medium contains from 0.25 to 75 g/L of at least one complex nitrogen source selected from the group consisting of soy meal, corn steep liquor, brewer's spent grains, wet distillers grains (WDG), dried distillers grains with solubles (DDGS), yeast extract, peptone or mixtures thereof. The amount of complex nitrogen source is to be calculated in accordance with the above definitions and required nitrogen content of the fermentation medium.

[0049] Within another preferred embodiment of the present invention, the at least one filamentous fungus cell is a genetically modified filamentous fungus cell with the ability to express at least one heterologous hydrolyase or oxidoreductase enzyme, such as but not limited to an enzyme belonging to the class of cellulases, belonging to the class of beta-glucosidases or belonging to the class of xylanases or belonging to the class of lytic polysaccharide monooxy-

genases. Within such a preferred embodiment, the at least one heterologous hydrolase or oxidoreductase enzyme preferably originates from another filamentous fungus such as—but not limited to—*Acremonium*, *Aspergillus*, *Chaetomium*, *Emericella*, *Fusarium*, *Humicola*, *Hypocrea*, *Irpex*, *Magnaporthe*, *Myceliophthora*, *Neurospora*, *Penicillium*, *Rhizopus*, *Talaromyces*, *Trichoderma* and *Trametes*. Within a particularly preferred embodiment the at least one filamentous fungus cell is a *Trichoderma reesei* cell and the at least one heterologous hydrolase or oxidoreductase enzyme originates from *Acremonium*, *Ajellomyces*, *Alternaria*, *Armillaria*, *Arthroderma*, *Aspergillus*, *Bionectria*, *Bipolaris*, *Ceriporiopsis*, *Chaetomium*, *Cladophialophora*, *Clohesyomyces*, *Colletotrichum*, *Coniochaeta*, *Coniosporium*, *Diaporthe*, *Dothistroma*, *Emericella*, *Epicoccum*, *Exophiala*, *Fomes*, *Fonsecaea*, *Fusarium*, *Gibberella*, *Grosmannia*, *Hebeloma*, *Hortaea*, *Humicola*, *Hypocrea*, *Hypoxyton*, *Irpex*, *Isaria*, *Kuraishia*, *Leucoagaricus*, *Madurella*, *Magnaporthe*, *Marssonina*, *Metarhizium*, *Moniliophthora*, *Myceliophthora*, *Mycosphaerella*, *Neurospora*, *Oidiendron*, *Ophiostoma*, *Paecilomyces*, *Paraphaeosphaeria*, *Penicillium*, *Phanerochaete*, *Phialophora*, *Pleurotus*, *Pochonia*, *Pseudocercospora*, *Pseudogymnoascus*, *Pyrenophora*, *Rasamsonia*, *Rhinoctadiella*, *Rhizopus*, *Rhizosphaera*, *Rhynchosporium*, *Setosphaeria*, *Sphaerulina*, *Sporothrix*, *Stachybotrys*, *Stemphylium*, *Talaromyces*, *Termitomyces*, *Tilletiaria*, *Torrubiella*, *Trametes*, *Trichoderma*, *Trichophyton*, *Uncinocarpus* and/or *Valsa* species.

[0050] According to the present invention, the at least one filamentous fungus cell as is a filamentous fungus cell wherein SEQ ID NO: 1 has been disrupted. The “disruption” can thereby be carried out by any means and measure known to the person skilled in the art as suitable for the purpose of disruption. The term “disruption” comprises all techniques that either lead to the gene no longer being transcribed or to the protein encoded by the gene no longer being produced or only being produced in an inactive form.

[0051] Exemplary methods which can be used within the present invention are:

[0052] the partial or complete removal from the genome of the gene, the region coding for the protein and/or the promoter or other regions necessary for the expression of the gene (=“deletion”)

[0053] the alteration of the DNA sequence of the coding region so that a shortened protein (=generation of a stop codon) and/or a protein with an altered amino acid sequence is produced which can no longer perform the function of the unchanged protein (=“mutation”)

[0054] the modification of the DNA sequence of the promoter or other regions necessary for the expression of the gene, so that the gene is no longer transcribed (=no RNA and therefore no protein is produced)

[0055] the expression of RNA with a sequence complementary to that of the target gene. This leads to hybridization (=pairing of complementary sequences) of the two RNAs and to a degradation of this double-stranded RNA. As a result, no RNA of the target gene is available for protein synthesis (=RNA interference).

[0056] Within the present invention SEQ ID NO:1 is defined within the sequence protocol.

[0057] Mixing according to step (c) of the inventive process of the present invention is carried out for a time period from 1 minute to 10 days, preferably from 10 hours to 7 days, further preferred from 24 hours to 5 days,

preferably under constant stirring with a power input from 150 to 20000 W/m³ and more preferably from 500 to 15000 W/m³ and under oxygen controlled conditions. The average dissolved oxygen level is preferably selected from 0.01% to 80%, preferred from 0.1% to 50%, particularly preferred from 5% to 30% and most preferred from 12% to 28%. Within a particularly preferred embodiment, the dissolved oxygen level is controlled by a stirrer or compressed air flow or internal reactor pressure or a combination of two or three of these measures. Furthermore, mixing according to step (c) of the inventive process is carried out at a temperature of from 20 to 35° C., preferably at a temperature of from 21 to 34° C., wherein a temperature selected from the range of from 22 to 33° C. is also preferred.

[0058] “Mixing” according to step (c) of the process of the present invention is preferably conducted in a batch mode (discontinuous), in a fed-batch mode or in a continuous mode. Most preferably, the inventive process is conducted in a fed-batch mode.

[0059] “Obtaining” according to step (d) of the inventive process is preferably carried out by harvesting the technical enzyme composition at the end of the time period applied for mixing during step (c) as it is without further treatment.

[0060] Within another preferred embodiment of the present invention, the inventive process further contains the step (e): subjecting the technical enzyme composition according to step d) to a purification method. The purification according to step (e) can be carried out by any measure known to a person skilled in the art as suitable for the inventive purpose. Suitable purification methods are selected from the group consisting of filtration (ultrafiltration, microfiltration, nanofiltration, depth filtration, sterile filtration, filter press), centrifugation, decantation, flotation, chromatographic separation, adsorption, electrodialysis, extraction, precipitation, crystallisation, spray drying, granulation, coating, extrusion or combinations thereof. Preferred are filter-based solid-liquid separations. It is further particularly preferred to use a filter press. The residues after the filtration should have a minimal solid content of 20% (wt./wt.), preferably 25% (wt./wt.), particularly preferred 30% (wt./wt.) and most preferred 40% (wt./wt.) solid content. In case the process according to the present invention involves solid-liquid separation as purification, the technical enzyme composition obtained according to step (d) of the inventive process is considered to be the liquid fraction.

[0061] Within a preferred embodiment of the inventive process, the process further comprises step

[0062] (ai) sterilization of the fermentation medium according to step (a).

[0063] Sterilization can thereby be carried out by any means or measure known to a person skilled in the art as suitable for the inventive purpose. Within a preferred embodiment, sterilization is carried out by filtration, such as but not limited to membrane filtration processes or by ultra high temperature heating. A combination of two or more sterilization methods is also possible, however, it is particularly preferred to only apply ultra high temperature heating (also referred to as UHT). The UHT treatment is preferably carried out at a temperature of from 100 to 155° C. and for a duration of from 10 to 30 seconds, more preferred at a temperature of from 120 to 140° C. for a duration of from 10 to 20 seconds.

[0064] Within another aspect, the present invention relates to a filamentous fungus cell wherein SEQ ID NO: 1 has been

disrupted. Disruption of SEQ ID NO: 1 can be carried out by any means and measure known to a person skilled in the art to be suitable for the inventive purpose. Possible and preferred methods and measures have been defined within the description. Within a preferred embodiment, SEQ ID NO: 1 has been disrupted by deletion, mutation, modification of a promotor or any other regulatory sequence, generation of a stop codon or RNA interference. The term “filamentous fungus cell” has been defined within the description. All definitions given apply.

[0065] Within a preferred embodiment, the filamentous fungus cell is a genetically modified filamentous fungus cell with the ability to express at least one heterologous hydrolyase enzyme. Such genetically modified filamentous fungus cell has been defined within the description. Within a particularly preferred embodiment of the present invention, the filamentous fungus cell is a genetically modified filamentous fungus cell wherein the filamentous fungus cell comprises at least one heterologous beta-glucosidase enzyme encoding sequence, at least one heterologous cellulase enzyme encoding sequence, at least one heterologous xylanase enzyme encoding sequence, at least one heterologous beta-xylosidase enzyme encoding sequence, at least one heterologous pectinase enzyme encoding sequence, at least one heterologous oxidase encoding sequence, at least one heterologous protease enzyme encoding sequence, at least one heterologous isomerase enzyme encoding sequence and/or at least one heterologous lytic polysaccharide monooxygenase enzyme encoding sequence.

[0066] In another aspect the present invention relates to a technical enzyme composition produced according to the process as defined before.

[0067] In a further aspect the present invention relates to the use of a filamentous fungus cell as defined before for the production of a technical enzyme composition as defined before.

Generally Preferred Embodiments

[0068] In the following, generally preferred embodiments of the present invention are listed which do not limit the scope of the invention and/or scope of the claims in any respect. The generally preferred embodiments illustrate particularly suitable embodiments for the production of technical enzyme composition by the filamentous fungus *Trichoderma reesei*.

Generally Preferred Embodiment 1

[0069] Process for production of a technical enzyme composition, comprising the following steps:

- [0070]** (a) providing a fermentation medium with a glucose content of from 5 to 550 g/L or from 5 to 450 g/L;
- [0071]** (b) addition of at least one *Trichoderma reesei* cell wherein SEQ ID NO:1 has been disrupted;
- [0072]** (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.;
- [0073]** (d) obtaining a technical enzyme composition.

Generally Preferred Embodiment 2

[0074] Process for production of a technical enzyme composition, comprising the following steps:

- [0075]** (a) providing a fermentation medium with a glucose content of from 5 to 550 g/L or from 5 to 450 g/L and wherein the fermentation medium further contains xylose and wherein the glucose to xylose ratio is selected from the range of from 1 to 3.5;
- [0076]** (b) addition of at least one *Trichoderma reesei* cell wherein SEQ ID NO:1 has been disrupted;
- [0077]** (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.;
- [0078]** (d) obtaining a technical enzyme composition.

Generally Preferred Embodiment 3

[0079] Process for production of a technical enzyme composition, comprising the following steps:

- [0080]** (a) providing a fermentation medium with a glucose content of from 5 to 550 g/L or from 5 to 450 g/L and wherein the fermentation medium is free from cellulose, hemicellulose, gluco-oligosaccharides and/or sophorose;
- [0081]** (b) addition of at least one *Trichoderma reesei* cell wherein SEQ ID NO:1 has been disrupted;
- [0082]** (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.;
- [0083]** (d) obtaining a technical enzyme composition.

Generally Preferred Embodiment 4

[0084] Process for production of a technical enzyme composition, comprising the following steps:

- [0085]** (a) providing a fermentation medium with a glucose content of from 5 to 550 g/L or from 5 to 450 g/L and a cellulose content of from 0.01 g/L to 1 g/L;
- [0086]** (b) addition of at least one *Trichoderma reesei* cell wherein SEQ ID NO:1 has been disrupted;
- [0087]** (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.;
- [0088]** (d) obtaining a technical enzyme composition.

Generally Preferred Embodiment 5

[0089] Process for production of a technical enzyme composition, comprising the following steps:

- [0090]** (a) providing a fermentation medium with a glucose content of from 5 to 550 g/L or from 5 to 450 g/L;
- [0091]** (b) addition of at least one *Trichoderma reesei* cell wherein SEQ ID NO:1 has been disrupted by deletion, mutation, modification of a promotor or any other regulatory sequence, generation of a stop codon or RNA interference and wherein the filamentous fungus cell comprises at least one heterologous beta-glucosidase enzyme encoding sequence, at least one heterologous cellulase enzyme encoding sequence, at least one heterologous xylanase enzyme encoding sequence, at least one heterologous beta-xylosidase

enzyme encoding sequence, at least one heterologous pectinase enzyme encoding sequence, at least one heterologous oxidase encoding sequence, at least one heterologous protease enzyme encoding sequence, at least one heterologous isomerase enzyme encoding sequence and/or at least one heterologous lytic polysaccharide monooxygenase enzyme encoding sequence;

[0092] (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.;

[0093] (d) obtaining a technical enzyme composition.

Generally Preferred Embodiment 6

[0094] Process for production of a technical enzyme composition, comprising the following steps:

[0095] (a) providing a fermentation medium with a glucose content of from 5 to 550 g/L or from 5 to 450 g/L and wherein the fermentation medium is free from sophorose and/or gluco-oligosaccharides;

[0096] (b) addition of at least one *Trichoderma reesei* cell wherein SEQ ID NO:1 has been disrupted by deletion, mutation, modification of a promotor or any other regulatory sequence, generation of a stop codon or RNA interference and wherein the filamentous fungus cell comprises at least one heterologous beta-glucosidase enzyme encoding sequence, at least one heterologous cellulase enzyme encoding sequence, at least one heterologous xylanase enzyme encoding sequence, at least one heterologous beta-xylosidase enzyme encoding sequence, at least one heterologous pectinase enzyme encoding sequence, at least one heterologous oxidase encoding sequence, at least one heterologous protease enzyme encoding sequence, at least one heterologous isomerase enzyme encoding sequence and/or at least one heterologous lytic polysaccharide monooxygenase enzyme encoding sequence;

[0097] (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.;

[0098] (d) obtaining a technical enzyme composition.

Generally Preferred Embodiment 7

[0099] Process for production of a technical enzyme composition, comprising the following steps:

[0100] (a) providing a fermentation medium with a glucose content of from 5 to 550 g/L or from 5 to 450 g/L wherein the fermentation medium can at least partly originate from chemical, mechanical and/or enzymatic hydrolysis of lignocellulosic biomass;

[0101] (b) addition of at least one *Trichoderma reesei* cell wherein SEQ ID NO:1 has been disrupted;

[0102] (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.;

[0103] (d) obtaining a technical enzyme composition.

Generally Preferred Embodiment 8

[0104] *Trichoderma reesei* cell, wherein SEQ ID NO: 1 has been disrupted by deletion, mutation, modification of a promotor or any other regulatory sequence, generation of a stop codon or RNA interference, comprising at least one heterologous beta-glucosidase enzyme encoding sequence, at least one heterologous cellulase enzyme encoding sequence, at least one heterologous xylanase enzyme encoding sequence, at least one heterologous beta-xylosidase enzyme encoding sequence, at least one heterologous pectinase enzyme encoding sequence, at least one heterologous oxidase encoding sequence, at least one heterologous protease enzyme encoding sequence, at least one heterologous isomerase enzyme encoding sequence and/or at least one heterologous lytic polysaccharide monooxygenase enzyme encoding sequence.

Generally Preferred Embodiment 9

[0105] *Trichoderma reesei* cell, wherein SEQ ID NO:1 has been disrupted by deletion, mutation, modification of a promotor or any other regulatory sequence, generation of a stop codon or RNA interference, comprising at least one heterologous beta-glucosidase enzyme encoding sequence, at least one heterologous cellulase enzyme encoding sequence, at least one heterologous xylanase enzyme encoding sequence, at least one heterologous beta-xylosidase enzyme encoding sequence, at least one heterologous pectinase enzyme encoding sequence, at least one heterologous oxidase enzyme encoding sequence, at least one heterologous protease enzyme encoding sequence, at least one heterologous isomerase enzyme encoding sequence and/or at least one heterologous lytic polysaccharide monooxygenase enzyme encoding sequence and wherein the at least one heterologous enzyme sequence originates from *Acremonium*, *Aspergillus*, *Chaetomium*, *Emericella*, *Fusarium*, *Humicola*, *Hypocrea*, *Irpex*, *Magnaporthe*, *Myceliophthora*, *Neurospora*, *Penicillium*, *Rhizopus*, *Talaromyces*, *Trichoderma* and *Trametes*.

Generally Preferred Embodiment 10

[0106] Technical enzyme composition produced according to a process as defined by any of generally preferred embodiments 1 to 7.

Generally Preferred Embodiment 11

[0107] Use of a filamentous fungus cell as defined by any of generally preferred embodiments 8 or 9 for the production of a technical enzyme composition.

Generally Preferred Embodiment 12

[0108] Process for production of a technical enzyme composition as defined by any of generally preferred embodiments 1 to 7, wherein the growth medium contains from 0.05 to 50 g/L nitrogen added in form of at least one complex nitrogen source selected from the group consisting of soy meal, corn steep liquor, brewer's spent grains, wet distillers grains (WDG), dried distillers grains with solubles (DDGS), yeast extract, peptone or mixtures thereof.

Generally Preferred Embodiment 13

[0109] Process for production of a technical enzyme composition as defined by any of generally preferred embodi-

ments 1 to 7 and 12, wherein the growth medium contains from 0.05 to 2 g/L nitrogen added in form of at least one complex nitrogen source selected from the group consisting of soy meal, corn steep liquor, brewer's spent grains, wet distillers grains (WDG), dried distillers grains with solubles (DDGS), yeast extract, peptone or mixtures thereof and wherein the fermentation medium is in the range of from 0.1 to less than 100 L.

Generally Preferred Embodiment 14

[0110] Process for production of a technical enzyme composition as defined by any of generally preferred embodiments 1 to 7 and 12, wherein the growth medium contains from 2 to 50 g/L nitrogen added in form of at least one complex nitrogen source selected from the group consisting of soy meal, corn steep liquor, brewer's spent grains, wet distillers grains (WDG), dried distillers grains with solubles (DDGS), yeast extract, peptone or mixtures thereof and wherein the fermentation medium is in the range of from 100 to 10000000 L.

FIGURES AND EXAMPLES

[0111] The present invention is described by the following figures and examples. It is thereby emphasized that the figures and examples do not limit the scope of the invention and claims but merely constitute further illustration of the invention, inventive purpose and benefits achieved by the inventive method.

LIST OF FIGURES

[0112] FIG. 1: Protein concentrations in the culture supernatants of pSEQ1M-HygR transformants MSEQ1-1 to -4 and reference strain M18.2b grown in shake flasks in medium 1. Values are given in relation to the protein concentration in the supernatants of the host strain M18.2b which is set to 1.

[0113] FIG. 2: Biomass concentrations in the culture broths of pSEQ1M-HygR transformants MSEQ1-1 to -4 and reference strain M18.2b grown in shake flasks in medium 1. Values are given in relation to the biomass concentration in the culture broth of the host strain M18.2b which is set to 1.

[0114] FIG. 3: Viscosity of culture broths of pSEQ1M-HygR transformants MSEQ1-1 to -4 and reference strain M18.2b grown in shake flasks in medium 1. Values are given in relation to the viscosity of the culture broth of the host strain M18.2b which is set to 1.

[0115] FIG. 4: SDS-PAGE gel of culture supernatants of pSEQ1M-HygR transformants MSEQ1-1 to -4 and reference strain M18.2b grown in shake flasks in medium 1.

[0116] FIG. 5: Protein concentrations in the culture supernatants of pSEQ1M-HygR transformants MSEQ1-1 to -4 and reference strain M18.2b grown in shake flasks in medium 2. Values are given in relation to the protein concentration in the supernatants of the host strain M18.2b which is set to 1.

[0117] FIG. 6: Biomass concentrations in the culture broths of pSEQ1M-HygR transformants MSEQ1-1 to -4 and reference strain M18.2b grown in shake flasks in medium 2. Values are given in relation to the biomass concentration in the culture broth of the host strain M18.2b which is set to 1.

[0118] FIG. 7: Viscosity of culture broths of pSEQ1M-HygR transformants MSEQ1-1 to -4 and reference strain M18.2b grown in shake flasks in medium 2. Values are given

in relation to the viscosity of the culture broth of the host strain M18.2b which is set to 1.

[0119] FIG. 8: SDS-PAGE gel of culture supernatants of pSEQ1M-HygR transformants MSEQ1-1 to -4 and reference strain M18.2b grown in shake flasks in medium 2.

GENERAL

[0120] The examples describe a way to disrupt the *Trichoderma reesei* SEQ1 gene by deleting a nucleotide resulting in a frame shift and consequently in a truncation of the encoded protein. They also show the effect of the SEQ1 gene disruption on the protein production, biomass formation and culture broth viscosity of *T. reesei*.

Example 1: Construction of a SEQ1 Mutation Vector

[0121] Standard methods known to those skilled in the art and described e.g. by Sambrook and Russel (Molecular Cloning—A laboratory manual; Cold Spring Harbor Laboratory Press, New York) or by Jansohn et al. (Gentechnische Methoden, Elsevier, München) were used for DNA agarose gel electrophoresis, purification of DNA, transformation of *Escherichia coli*, plasmid propagation and purification, amplification of pieces of DNA by polymerase chain reaction (PCR) and isolation of genomic DNA from *Trichoderma reesei*. Ligation-independent cloning (LIC) was done essentially as described by Aslanidis and de Jong (1990, Nucleic Acid Res. 18 (20), 6069). All restriction enzymes were purchased from New England Biolabs and used according to the manufacturer's instructions. Purification of restriction digested, PCR-amplified and gel purified DNA was done using the Wizard® SV Gel and PCR Clean-Up System from Promega.

[0122] A SEQ1 mutation vector was constructed by fusing the Hygromycin B resistance marker to the SEQ1 5' and 3' flanking regions and cloning the fusion product in a pUC19-derived plasmid. The flanking regions contain a part of the SEQ1 coding region that introduces a mutation encompassing the deletion of the nucleotide C1755 (position according to SEQ ID NO: 1) into the SEQ1 gene.

[0123] The SEQ1 5' flanking region (ca. 2.6 kb) was amplified from genomic DNA from *Trichoderma reesei* M18.2b (DSM 19984) as a template using the primers SEQ1fl5fw (5'-AACGCCTTTCCTGTATCGTC-3'; SEQ ID NO: 2) and SEQ1fl5rv (5'-TTGATCGCGTCAGCTTGTCGAATCTCCTC-CAC TAGTGCAAAGATCCTGGCAAGC-3'; SEQ ID NO: 3) and phusion polymerase from Thermo Scientific according to the manufacturer's instructions (annealing temperature: 63.4° C., elongation time: 1 min 20 sec, 30 cycles).

[0124] The SEQ1 3' flanking region (ca. 2.5 kb) was amplified from genomic DNA from *Trichoderma reesei* M18.2b (DSM 19984) as a template using the primers SEQ1fl3fw (5'-TCAGCTCTATTGGCTTGCCAGGATCTTTGCACTAGTGGAGGAGATTCGACAAGC TG-3'; SEQ ID NO: 4) and SEQ1fl3rv (5'-ATGTGTTGCTCAAGTGATGC-3'; SEQ ID NO: 5) and phusion polymerase from Thermo Scientific according to the manufacturer's instructions (annealing temperature: 62.4° C., elongation time: 1 min 20 sec, 30 cycles).

[0125] The PCR-amplified SEQ1 5' and 3' flanking region were purified and fused using phusion polymerase from Thermo Scientific and the primers fus1 (5'-AAACCA-

GACAGACAGTCCTGCAGGCTCATCTGCTCT-CATGGGTG-3'; SEQ ID NO: 6) and fus2 (5'-AGAGAG-GAGAGACAGTCCTGCAGGGCTACAGTTGGCAAGATGTTTC-3'; SEQ ID NO: 7). Approximately 100 ng of both templates and 20 U of primers fus1 and fus2, respectively, were used. The PCR consisted of 10 initial cycles of 10 sec at 98° C., 30 sec at 68° C. and 2 min 15 sec at 72° C. followed by cooling to 10° C. Then the primers were added, followed by a 30 sec hold at 98° C. and 30 cycles of 10 sec at 98° C., 30 sec at 62.7° C. and initially 1 min 45 sec at 72° C. with the 72° C. incubation being extended by 5 sec per cycle. The PCR was concluded by a 10 min hold at 72° C. and cooling to 10° C.

[0126] The approx. 5.0 kb long fusion PCR product was purified and cloned into a PshAI-linearized pUC19-derived plasmid (SEQ ID NO: 8) that contained a LIC reception site instead of the multiple cloning site. The linearized vector was treated with T4 DNA polymerase in the presence of dTTP. The fusion PCR product was treated with T4 DNA polymerase in the presence of dATP. T4 DNA polymerase treated vector and fusion PCR amplicon were mixed and annealed as described by Aslanidis and de Jong. The LIC assay was then transformed in chemically competent *Escherichia coli* XL1-Blue cells (Agilent), plated on LB-Agar plates containing 100 mg·l⁻¹ ampicillin (LB-Amp) and incubated at 37° C. for 24 h. Colonies were picked from the agar plates using toothpicks, transferred into liquid LB-Amp medium and incubated at 37° C. for 24 h with shaking (250 RPM). Plasmid DNA was isolated and integration of the insert was verified by digestion with SpeI. Plasmid clones were verified by Sanger sequencing using primers 53SEQ-1 (5'-TCATGAGCGGATACATATTTG-3'; SEQ ID NO: 9), 53SEQ-2 (5'-TTTTGCGATGATGGCCTAG-3'; SEQ ID NO: 10), 53SEQ-3 (5'-CAAAGACTCCAAAGACGAGC-3'; SEQ ID NO: 11), 53SEQ-4 (5'-TGCTAGATGAACAGATCGGC-3'; SEQ ID NO: 12) and 53SEQ-5 (5'-GTCATGGAGGATTACAGGC-3'; SEQ ID NO: 13), and one plasmid with the correct sequence was designated pSEQ1-5-3

[0127] In order to introduce a LIC site into pSEQ1-5-3, the plasmid was linearized by digestion with SpeI and purified. Then 1 µl each of 10 U solutions of oligonucleotides LICfw (5'-CTAGGTAACAA-GACACAGCCCCGGGCTCTTGTCTGTTAC-3'; SEQ ID NO: 14) and LICrv (5'-CTAGGTAACAGACAAGAGCCCCGGGCTGTGTCTTGTGTTAC-3'; SEQ ID NO: 15) were mixed in a PCR tube, placed in 70° C. warm water and let cool down to room temperature (duration: ca. 2 h). After cooling down, the LICfw-LICrv-mixture was ligated with SpeI-digested pSEQ1-5-3 by mixing the 2 µl of LICfw-LICrv mixture, 3 µl of purified SpeI-digested pSEQ1-5-3 (ca. 100 ng of plasmid DNA), 1 µl of 10× T4 Ligase Puffer (Promega), 1 µl of PEG solution (500 g·l⁻¹ Polyethylene glycol 3350 dissolved in nuclease-free water), 1 µl of T4 DNA Ligase (Promega) and 2 µl of nuclease-free water and incubating the mixture at 20° C. for 1 h. The DNA was purified using the Wizard® SV Gel and PCR Clean-Up System (Promega) and eluted with 50 µl of nuclease-free water. Then 6 µl of 10× T4 DNA Polymerase buffer were added to the purified DNA solution, and the volume of the mixture was adjusted to 60 µl by addition of nuclease-free water. The tube with the 60 µl of mixture was then placed in a beaker with boiling water and let cool down to room temperature (duration: ca. 3 h). The DNA was then used to

transform chemically competent *Escherichia coli* XL1-Blue cells (Agilent). The transformants were plated on LB-Agar plates containing 100 mg·l⁻¹ ampicillin (LB-Amp) and incubated at 37° C. for 24 h. Colonies were picked from the agar plates using toothpicks, transferred into liquid LB-Amp medium and incubated at 37° C. for 24 h with shaking (250 RPM). Plasmid DNA was isolated and integration of the insert was verified by digestion with SrfI. Plasmid clones were verified by Sanger sequencing using primer 53SEQ-5 (5'-GTCATGGAGGATTACAGGC-3'; SEQ ID NO: 13) and one plasmid with the correct sequence was designated pSEQ1-5-3-LIC.

[0128] The Hygromycin B resistance marker cassette (SEQ ID NO: 16) had been synthesized by Thermo Scientific. Primers hygrfw (5'-AACAAGACACAGCCC-TATAAC-3'; SEQ ID NO: 17) and hygrv (5'-AACA-GACAAGAGCCCTATAAC-3'; SEQ ID NO: 18) were used to amplify the approximately 2.4 kb long cassette (annealing temperature: 60.3° C., elongation time: 40 sec, 30 cycles) using phusion polymerase from Thermo Scientific according to the manufacturer's instructions. The SrfI-linearized vector pSEQ1-5-3-LIC was treated with T4 DNA polymerase in the presence of dTTP. The PCR-amplified Hygromycin B resistance marker cassette was treated with T4 DNA polymerase in the presence of dATP. T4 DNA polymerase treated vector and insert were mixed and annealed as described in by Aslanidis and de Jong. The assay was then transformed in chemically competent *Escherichia coli* XL1-Blue cells (Agilent), plated on LB-Agar plates containing 100 mg·l⁻¹ ampicillin (LB-Amp) and incubated at 37° C. for 24 h. Colonies were picked from the agar plates using toothpicks, transferred into liquid LB-Amp medium and incubated at 37° C. for 24 h with shaking (250 RPM). Plasmid DNA was isolated and integration of the insert was verified by digestion with SbfI. Plasmid clones were verified by Sanger sequencing using primers 53SEQ-1 (5'-TCATGAGCGGATACATATTTG-3'; SEQ ID NO: 9), 53SEQ-2 (5'-TTTTGCGATGATGGCCTAG-3'; SEQ ID NO: 10), 53SEQ-3 (5'-CAAAGACTCCAAAGACGAGC-3'; SEQ ID NO: 11), 53SEQ-4 (5'-TGCTAGATGAACAGATCGGC-3'; SEQ ID NO: 12) and 53SEQ-5 (5'-GTCATGGAGGATTACAGGC-3'; SEQ ID NO: 13), FullSEQ-1 (5'-GGCG-GAGCCTATGGAAAAAC-3'; SEQ ID NO: 19), FullSEQ-2 (5'-TCCTCCTCCTACTCTCCATC-3'; SEQ ID NO: 20), FullSEQ-3 (5'-GCTGGTATTGGTCATGTAGC-3'; SEQ ID NO: 21), FullSEQ-4 (5'-GTTGGCCCAGAAACATCC-3'; SEQ ID NO: 22), FullSEQ-5 (5'-AGATCCTATTGACCTCTCTGC-3'; SEQ ID NO: 23), FullSEQ-6 (5'-CCCAGACCACCTGCACACTC-3'; SEQ ID NO: 24), FullSEQ-7 (5'-GCAAGACCTGCCTGAAAC-3'; SEQ ID NO: 25), FullSEQ-8 (5'-CTGGACCGATGGCTGTGTAG-3'; SEQ ID NO: 26) and FullSEQ-9 (5'-GG-GAGAGAAATCAGCAGGTG-3'; SEQ ID NO: 27) and one plasmid with correct sequence was designated pSEQ1M-HygR.

Example 2: Transformation of the SEQ1 Mutation Vector into *Trichoderma reesei*

[0129] Vector pSEQ1M-HygR was digested with SbfI according to the manufacturer's instructions and the mutation cassette (7.4 kb) was purified by agarose gel electrophoresis and with the Wizard PCR purification kit from Promega. *Trichoderma reesei* M18.2b (DSM 19984) was transformed with the digested vector essentially as described

in Penttilä et al (1987) Gene 61:155-164. The transformants were selected on potato dextrose agar plates containing 100 mg·l⁻¹ of Hygromycin B and 1 M sorbitol and purified by singularisation. Conidia stocks of the purified strains were prepared by growing them on potato dextrose agar plates at 30° C. until the plates were covered with spores. The conidia were harvested with sterile sodium chloride (0.9 g·l⁻¹)-Triton X-100 (0.01 g·l⁻¹) solution, adjusted to OD600=10 with sterile water, supplemented with glycerol to a final concentration of 50 g·l⁻¹ and stored at -80° C.

[0130] Genomic DNA was isolated from the mycelium of the transformants and the host strain. The integration of the SEQ1 mutation cassette at the intended locus was verified by PCR using phusion polymerase from Thermo Fisher Scientific according to the manufacturer's instructions, genomic DNA from the transformants as template and primers SEQ1MKO1fw (5'-GCATTGAGTTGAGCGCTAAC-3'; SEQ ID NO: 28) and SEQ1MKOrv (5'-CCATGGTTCGAACGGAAAC-3'; SEQ ID NO: 29) (annealing temperature: 61.8° C., elongation time: 55 sec, 30 cycles) or primers SEQ1MKO2fw (5'-TGTATCAAGCTAGGTGGGAG-3'; SEQ ID NO: 30) and SEQ1MKOrv (5'-CCATGGTTCGAACGGAAAC-3'; SEQ ID NO: 29) (annealing temperature: 61.5° C., elongation time: 55 sec, 30 cycles), respectively. A 2.7 kb band with primers SEQ1MKO1fw and SEQ1MKOrv indicates the integration of the mutation cassette at the SEQ1 locus, while a 2.6 kb band with primers SEQ1MKO2fw and SEQ1MKOrv indicates that the SEQ1 locus is still native (i.e. this band is not expected with genomic DNA from transformants that had integrated the pSEQ1M-HygR fragment at the intended locus). Genomic DNA from strain M18.2b was also tested as a control. In order to verify that the intended mutation had been inserted into the SEQ1 ORF, the amplicon obtained with primers SEQ1MKO1fw and SEQ1MKOrv was sequenced using primer M1Seq-01 (5'-GCCAATAGAGCTGAGAAGTG-3'; SEQ ID NO: 31) and M1Seq-02 (5'-TCTGAAGAGGGCTGAGAAAG-3'; SEQ ID NO: 32).

[0131] Four transformants containing the mutation from pSEQ1M-HygR in the SEQ1 ORF were named MSEQ1-1 to -4.

Example 3: Growth of the SEQ1 Deletion Strains in Shake Flasks

[0132] The strains MSEQ1-1 to -4 and M18.2b were grown in shake flasks in medium 1 and in medium 2. Medium 1 contains (g·l⁻¹):

Name	Concentration [g/l]
Acetic acid	0.34
Calcium	0.12
Chloride, water soluble	0.15
Copper	0.0001
Fat (HCl soluble)	0.001
Furfural	0.003
Glucose	6.5
Glycerol	0.009
HMF	0.006
Iron	0.004
Magnesium	0.048
Manganese	0.002
Na-D/L-Lactat	0.097

-continued

Name	Concentration [g/l]
Nitrogen, soluble	0.85
Phosphorus	0.48
Phthalate	8.2
Potassium	3.2
Sodium	0.015
Sulfur	0.86
Xylose	3.6
Zinc	0.001

[0133] The medium was adjusted to pH 5.5 with HCl or NaOH and sterilized by autoclaving (20 min at 121° C.).

[0134] Medium 2 contains (g·l⁻¹):

Name	Concentration [g/l]
(NH4) ₂ SO ₄	2.8
KH ₂ PO ₄	2.0
FeSO ₄ × 7H ₂ O	0.02
MnSO ₄ × H ₂ O	0.0064
ZnSO ₄ × 7H ₂ O	0.0056
CuSO ₄ × 5H ₂ O	0.0004
Bacto™ Yeast Extract, technical (Thermo Fisher Scientific)	0.5
Glucose	10
CaCl ₂ × 2H ₂ O	0.3
MgSO ₄ × 7H ₂ O	0.3

[0135] The medium was adjusted to pH 5.5 with HCl or NaOH and sterilized by autoclaving (20 min at 121° C.).

[0136] 15 ml of the media were distributed into 50 ml Erlenmeyer shake flasks under a sterile hood. Conidia stocks of strains MSEQ1-1 to -4 and M18.2b were thawed, 75 µl of the conidia suspensions were pipetted into the Erlenmeyer flasks with the medium under a sterile hood and the flasks were closed with rubber foam caps. At least three flasks were inoculated per strain. The flasks were incubated at 30° C. with shaking (250 RPM) for 6 days. After 6 days, the cultures were poured into 15 ml tubes. Aliquots were removed, centrifuged (3220×g, 4° C., 15 min) and the supernatants stored at 4° C., while the remaining culture broth was used for determination of the biomass and viscosity (see below).

Example 4: Characterization of the Culture Supernatants and Broths: Protein Concentration, SDS-PAGE, Biomass, Viscosity

[0137] Protein concentrations in the centrifuged culture supernatants of strains MSEQ1-1 to -4 and M18.2b were measured using the Quick Start™ Bradford reagent (BioRad) and BSA standard solutions (BioRad) according to the supplier's instructions. The results of the measurements are shown in FIG. 1 and FIG. 5. Values are given in relation to the average protein concentration in the supernatants of the host strain M18.2b which is set to 1. It is obvious from these data that strains MSEQ1-1 to -4 produce significantly more protein than the host strain M18.2b.

[0138] For biomass determination, Whatman™ filter discs (P1) were dried at 60° C. until their weight remained constant for 24 h, cooled to room temperature and weighed. Culture broths of strains MSEQ1-1 to -4 and M18.2b were filtered using those dried filter discs and the mycelia were

washed with at least ten times the broth's volume of deionized water. Then the filter discs with the mycelia were dried at 60° C. until their weight remained constant for 24 h. The filter discs with the dried mycelia were weighed. The biomass concentration in the culture broth was then calculated by subtracting the mass of the dried filter disc from the mass of the dried filter disc with the mycelia and then dividing that value by the volume of the culture broth that had been filtered. The results of the measurements are shown in FIG. 2 and FIG. 6. Values are given in relation to the average biomass concentration in the supernatant of the host strain M18.2b which is set to 1. It is obvious from these data that strains MSEQ1-1 to -4 produce significantly less biomass than the host strain M18.2b.

[0139] The viscosity of the culture broths of strains MSEQ1-1 to -4 and M18.2b was measured using a Malvern Kinexus Lab+KNX2110 rotational rheometer with the Vane tool (4Vnn: CUPnn) according to the manufacturer's instructions. The measurements were taken at a temperature of 20° C. and at a rotation velocity of 18.11 RPM ("rotations per minute"). The viscosity values are depicted in FIG. 3 and FIG. 7 and are presented in relation to the viscosity of the culture broth of strain M18.2b, which is set to 1. It is obvious

from these data that the viscosity of the culture broths produced with MSEQ1-1 to -4 is significantly lower than that of the host strain M18.2b.

[0140] SDS-PAGE analysis of the centrifuged culture supernatants of strains MSEQ1-1 to -4 and M18.2b was done using methods known to those skilled in the art (e.g. described by Jansohn et al. (Gentechnische Methoden, Elsevier, München)) and the Criterion XT system (BioRad). Equal volumes of culture supernatants were loaded in each lane. Precision Plus Protein™ All Blue Standards (BioRad) was used as protein size reference. The gel images are shown in FIG. 4 and FIG. 8.

SUMMARY

[0141] Taken together these data demonstrate that the disruption of the SEQ1 gene results in a significantly more efficient protein production, with more protein and less biomass being formed, independent of the culture medium. In addition, the viscosity of the culture broth is significantly reduced as well.

SEQUENCE LISTING

SEQ ID NO: 1
 SEQ1 native gene
 ATGAGTAGAAACCGTCGAGAGTCCCAAAATATCTTGGAGTCACTGCATTCAAGG
 TACGCGTGATACGGCAAGTTCCACGGGCATATACAGCAAATTACCATATCCAA
 GTCCTTACATGGTAAACCAATAGCAACCCCTTTCTTTGCCCTTCAGGTTGTCCCG
 TCTCCGCGATGGTTTCAGGGTAAGGTGACGAACGGTACAGTAACAAAGACTCCA
 AAGACGAGCCACGATCGGTGGAAGCAATGCGTCATGCTGAGTCATCGACGCC
 CCTCCGTGGGTAAACAGGCAATGCCCGCCAAACAGCCGTGAGAAGCAAAATAA
 CATCATGACAGCTTCCAGCGCCTTGCTTTTGCTCTCCTGCACCGCTCCTCTCCC
 TCGTTGCAGCTATTTCGATTGTCTCTACTCGAGGCTCGACGGCGGCCCGCGGCC
 CAAAATGTCCACGTACGTGACGCTATCGTCTACATCTCTCTGACGCTCTATACCT
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 CACCTCACAGGCCAACCACCAATATCCTCAGCATCCCATTTTCGCCGGTCTGCTGC
 ACCTGTCTGCTGTGACGACGATCCGACAGTACATCAACACCAAATATGACCAGC
 ACCCGGACATGTTCCAGTATGACCTCGAGGCCATCGATGCGCTGCGCCGCGAC
 GCGGTGAACGTGCGCGAGCCGACCTGAGCGGCATCAAGAAGCTGCAGGTGT
 ACGCGGGCCAGCTGGTGTGGATTGGCGGCAAGTTTCCGATTGATGTGCGTAGA
 CGAAAGACGAGTAGGGGAGGAGCAGGAGAAACAAGCGGACAAGATGCTGAT
 GCTGCTAGATGAACAGATCGGCGCCGAGTTACCTGGTACCCGGCCCTTGGCT
 ACCACACCGACCGGCCGATGGCGCGCAACAACCTCAAGTACGAGCTCATGAAT
 GTCCTCTACAACCTCGCCGCCCTTGACTCTCAGCTTGCCCTCAACACGCCCCCG
 GGCGATACCGAGGGTCTCAAGTCCGCCGCCAACTACTTTTCCCTAGCCGCCCG
 CGTCTCTCCACATTCCAGAAAGCCGTGCTTCCCGAGCTGCGCATGTCCGACC
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 CGGTCTAGGAGGATTTACAGGCCCTTGAAGCGAAAGCTGGAAGAGGATCTGAAG
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 AACTTACGCAAAGATTGTCCCCAAAGCTGACAGCCACCAACAGATCCTATACCC
 CCAAAGGCAGAACTGAAGATCCTGGAGAGAGCCAACATGGCTGTTGCTCGAAC
 GCCCCCCCAGGTAGCCAATCCGCTTGACTACCTAGGTGACCATGCCGAGCTTG
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 TCTACGAGGAGCGCAGAGATCGGCTGGTCAACCAAACATCATTCAAGAGCTG
 GAGAACCTGACCGACAAGATCCACACACTTCTCAGCTCTATTGGCTTGCCAGGA
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SEQ ID NO: 2
SEQ1f15fw
AACGCCTTTCCTGTATCGTC

SEQ ID NO: 3
SEQ1f15rv
TTGATCGCGTCAGCTTGTCGAATCTCCTCCACTAGTGCAAAGATCCTGGCAAGC

SEQ ID NO: 4
SEQ1f13fw
TCAGCTCTATTGGCTTGCCAGGATCTTTGCACTAGTGGAGGAGATTTCGACAAGC

SEQ ID NO: 5
SEQ1f13rv
ATGTGTTGCTCAAGTGATGC

SEQ ID NO: 6
fus1
AAACCAGACAGACAGTCCTGCAGGCTCATCTGCTCTCATGGGTG

SEQ ID NO: 7
fus2
AGAGAGGAGAGACAGTCCTGCAGGGCTACAGTTGGCAAGATGTTT

SEQ ID NO: 8
LIC reception vector
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AAGCA

SEQ ID NO: 9
53SEQ-1
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SEQ ID NO: 10
53SEQ-2
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SEQ ID NO: 11
53SEQ-3
CAAAGACTCCAAAGACGAGC

SEQ ID NO: 12
53SEQ-4
TGCTAGATGAACAGATCGGC

SEQ ID NO: 13
53SEQ-5
GTCATGGAGGATTTACAGGC

SEQ ID NO: 14
LICfw
CTAGGTAACAAGACACAGCCCGGGCTCTTGTCTGTTAC

SEQ ID NO: 15
LICrv
CTAGGTAACAGACAAGAGCCCGGGCTGTGTCTTGTTC -3'

SEQ ID NO: 16
Hygromycin B resistance marker
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TGTCGTGT

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hygrfw
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SEQ ID NO: 18
hygrrv
AACAGACAAGAGCCCTATAAC

SEQ ID NO: 19
FullSEQ-1
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SEQ ID NO: 20
FullSEQ-2
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SEQ ID NO: 21
FullSEQ-3
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SEQ ID NO: 22
FullSEQ-4
GTTGGCCCAGAAACATCC

SEQ ID NO: 23
FullSEQ-5
AGATCCTATTGACCTCTCTGC

SEQ ID NO: 24
FullSEQ-6
CCCAGACCACCTGCACACTC

SEQ ID NO: 25
FullSEQ-7
GCAAGACCTGCCTGAAAC

SEQ ID NO: 26
FullSEQ-8
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SEQ ID NO: 27
FullSEQ-9
GGGAGAGAAATCAGCAGGTG

SEQ ID NO: 28
SEQ1MK01fw
GCATTGAGTTGAGCGCTAAC

SEQ ID NO: 29
SEQ1MK0rv
CCATGGTCGAACGGAAAC

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1. Process for production of a technical enzyme composition, comprising the following steps:
- (a) providing a fermentation medium with a glucose content of from 5 to 550 g/L;
 - (b) addition of at least one filamentous fungus cell wherein SEQ ID NO:1 has been disrupted;
 - (c) mixing of the fermentation medium and the at least one filamentous fungus cell for a time period of from 1 minute to 10 days at a temperature of from 20 to 35° C.; and
 - (d) obtaining a technical enzyme composition; wherein the at least one filamentous fungus cell is a *Trichoderma reesei* cell.
2. Process according to claim 1, wherein the pH of the fermentation medium according to step (a) has been adjusted to a pH selected from pH 2.0 to 6.0.
3. Process according to claim 1, wherein the fermentation medium further contains xylose and wherein the glucose to xylose ratio is selected from the range of from 1 to 3.5.
4. Process according to claim 1, wherein the fermentation medium further contains lactose and wherein the glucose to lactose ratio is selected from the range of from 1 to 10.
5. Process according to claim 1, wherein no glucoligosaccharides have been added to the fermentation medium.
6. Process according to claim 1, wherein no sophorose has been added to the fermentation medium.
7. Process according to claim 1, further comprising step (ai) sterilization of the fermentation medium according to step (a).
8. Process according to claim 1, wherein the fermentation medium has a potassium hydrogen phosphate content of

- from 0.5 to 10.0 g/L, a magnesium sulfate heptahydrate content of from 0.05 to 1 g/L, a calcium chloride dihydrate content of from 0.1 to 1 g/L, an ammonium sulfate content of from 1.5 to 4.5 g/L, an iron (II) sulfate heptahydrate content of from 0.005 to 0.1 g/L, a manganese sulfate content of from 0.00001 to 0.001 g/L, a zinc sulfate heptahydrate content of from 0.001 to 0.01 g/L and/or a copper sulfate pentahydrate content of from 0.0001 to 0.001.
9. Process according to claim 1, wherein the fermentation medium has a nitrogen content of from 0.05 to 50.0 g/L.
10. (canceled)
11. Process according to claim 1, wherein the filamentous fungus cell comprises at least one heterologous beta-glucosidase enzyme encoding sequence, at least one heterologous cellulase enzyme encoding sequence, at least one heterologous xylanase enzyme encoding sequence, at least one heterologous beta-xylosidase enzyme encoding sequence, at least one heterologous pectinase enzyme encoding sequence, at least one heterologous oxidoreductase encoding sequence, at least one heterologous protease enzyme encoding sequence, at least one heterologous isomerase enzyme encoding sequence and/or at least one heterologous lytic polysaccharide monooxygenase enzyme encoding sequence.
12. Process according to claim 1, further comprising the step
- (e) subjecting the technical enzyme composition according to step d) to a purification method.
13. Process according to claim 1, wherein the filamentous fungus cell is a *Trichoderma filamentous-reesei* fungus cell and wherein the fermentation medium contains from 0.05 to 50 g/L nitrogen

added in form of a complex nitrogen source selected from the group consisting of soy meal, corn steep liquor, brewer's spent grains, wet distillers grains (WDG), dried distillers grains with solubles (DDGS), yeast extract, peptone or mixtures thereof.

14. *Trichoderma reesei* filamentous fungus cell wherein SEQ ID NO:1 has been disrupted.

15. *Trichoderma reesei* filamentous fungus cell according to claim **14**, wherein SEQ ID NO:1 has been disrupted by deletion, mutation, modification of a promotor or any other regulatory sequence, generation of a stop codon or RNA interference.

16. Filamentous fungus cell according to claim **14**, wherein the at least one filamentous fungus cell is a genetically modified filamentous fungus cell, wherein the filamentous fungus cell comprises at least one heterologous beta-glucosidase enzyme encoding sequence, at least one

heterologous cellulase enzyme encoding sequence, at least one heterologous xylanase enzyme encoding sequence, at least one heterologous beta-xylosidase enzyme encoding sequence, at least one heterologous pectinase enzyme encoding sequence, at least one heterologous oxidoreductase encoding sequence, at least one heterologous protease enzyme encoding sequence, at least one heterologous isomerase enzyme encoding sequence, and/or at least one heterologous lytic polysaccharide monooxygenase enzyme encoding sequence.

17. Technical enzyme composition produced according to a process as defined in claim **1**.

18. Process according to claim **1**, wherein SEQ ID NO:1 has been disrupted by deletion, mutation, modification of a promotor or any other regulatory sequence, generation of a stop codon or RNA interference.

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